

PreOne Canonical Product Flow

Complete Flow Audit of the PreOne Documentation Set and the Single Source of Truth for Development

Sources audited: Master PRD, BRC, DDD, UI Philosophy, ERD v3, Prisma Schema v3, API Catalog, OpenAPI, ADR Series Volumes 1-4, Backend TD, Frontend Architecture, Security Architecture, Engineering Standards, Test Strategy, DevOps ADR-111 (19 documents)
Audience: CTO, Tech Lead, Product, UI/UX, Backend, Frontend, QA, DevOps
Status: Canonical reference - supersedes conflicting flows on adoption

Table of Contents

01. Executive Flow Verdict.....	1
1.1 Verdict by Layer.....	2
1.2 The Ten Canonical Decisions That Fix the Product.....	2
1.3 What Each Team Should Read First.....	2
02. Current Documentation Problems.....	2
2.1 What Is Already Correct (Preserve As-Is).....	3
2.2 What Is Inconsistent (Conflict Register).....	4
2.3 What Is Duplicated.....	4
2.4 What Is Missing.....	5
2.5 What Is Ambiguous.....	6
2.6 What Needs A Product Decision.....	7
03. Corrected Role Architecture.....	7
3.1 Canonical Role Hierarchy.....	7
3.2 Mapping Of Every Source Role Name.....	7
3.3 Per-Role Authority Profile.....	8
04. Corrected Tenant Architecture.....	8
4.1 Canonical Hierarchy.....	9
4.2 Where Everything Belongs.....	11
4.3 Multi-Tenant Isolation Contract.....	11
05. Authentication Architecture.....	12
5.1 Architecture Options Evaluated.....	13
5.2 Canonical Login Flow (Every Branch).....	13
5.3 First-Login, Invitation And Password Setup.....	18
5.4 Session Handling And Logout.....	18
5.5 Unauthorized, Suspended And Inactive Behaviours.....	19

5.6 Credential Policies.....	19
06. School Creation Flow.....	20
6.1 Canonical Lifecycle And Statuses.....	20
6.2 Creation Steps.....	21
6.3 Activation, Suspension And Offboarding.....	23
07. Onboarding Flow.....	24
08. Setup Flow.....	24
09. Settings Flow.....	25
10. Admissions Flow.....	25
10.1 The Five Status Domains.....	26
10.2 Canonical Pipeline.....	26
10.3 Duplicate And Edge Handling.....	29
11. Enrollment Flow.....	30
12. Student Lifecycle.....	34
12.1 Promotion, Transfer, Withdrawal, Archive, Deactivate, Delete.....	34
13. Parent Lifecycle.....	34
13.1 Canonical Parent Flow.....	35
13.2 Guardian Rules.....	37
14. Teacher / Staff Lifecycle.....	37
14.1 What Teachers Can And Cannot Approve.....	40
15. Attendance Flow.....	40
15.1 Canonical Marking Workflow.....	40
15.2 Statuses And Rules.....	43
16. Activities Flow.....	44
16.1 Permissions And Quality.....	44
17. Growth Passport Flow.....	45
17.1 Proposed Canonical Composition (RECOMMENDED).....	45
17.2 Permissions And Data Rules (RECOMMENDED).....	47
18. Fees & Finance Flow.....	48

18.1 Concepts And Lifecycle.....	48
18.2 Rulings Applied To Conflicts.....	51
19. Communication Flow.....	52
19.1 Who Can Message Whom.....	52
19.2 Channels And Delivery Pipeline.....	52
19.3 Event-Triggered Notification Catalog.....	53
20. AI Center Flow.....	53
20.1 Canonical AI Pipeline.....	54
20.2 Capability Catalog And Review Requirements.....	55
20.3 Classification Of Generated Content.....	56
20.4 Persistence And Guardrails.....	56
21. Reporting Flow.....	57
22. Audit Flow.....	58
23. Navigation Architecture.....	59
23.1 Final Information Architecture.....	60
23.2 Page-Level Specifications (Core Pages).....	63
24. RBAC Matrix.....	64
24.1 Concept Separation.....	64
24.2 Permission Grammar (resolves C21).....	64
24.3 Role x Module Action Matrix.....	65
24.4 Approval Authority Map.....	65
25. State Machines.....	65
25.1 School (Tenant).....	66
25.2 User (Staff / Owner / Parent).....	66
25.3 Admission Lead.....	67
25.4 Application.....	67
25.5 Document (per checklist item).....	68
25.6 Enrollment.....	69
25.7 Student.....	69

25.8 Parent.....	70
25.9 Attendance (day-state per student).....	71
25.10 Invoice / Payment.....	71
25.11 Growth Passport (RECOMMENDED).....	72
25.12 Notification.....	72
26. API Flow.....	73
26.1 Canonical Request Path.....	73
26.2 Response Envelope (resolves C19).....	74
26.3 Error Code Families (resolves C20).....	75
26.4 Conventions.....	75
27. Database Relationship Flow.....	76
27.1 Canonical Entity Catalog (Core).....	76
27.2 Cross-Cutting Conventions.....	77
28. Edge Cases.....	77
29. End-to-End Flow Diagrams.....	78
29.1 Platform & School Bootstrap.....	78
29.2 Admissions To Enrollment.....	79
29.3 Daily Operations & Learning.....	80
29.4 Finance.....	80
29.5 Communication, Insight & Reporting.....	81
29.6 Student & Staff Exits, Platform Actions.....	82
30. FINAL CANONICAL PREONE FLOW.....	82
31. Open Decisions / Items Requiring Product Approval.....	87
32. Development Readiness Checklist.....	88

Note: This Table of Contents is generated via field codes. To ensure page number accuracy after editing, please right-click the TOC and select "Update Field."

01. Executive Flow Verdict

This document is the single canonical source of truth for how PreOne works, produced by auditing the complete PreOne documentation set end to end: the Master PRD v1.0, Business Rules Catalog v1.0 (BRC), Domain-Driven Design v1.0, UI Design Philosophy v1.0, ERD v3.0, Prisma Schema v3.0, API Contract Catalog v1.0, OpenAPI Specification v1.0, the four ADR Series volumes (ADR-001 to ADR-090), Backend Technical Design v1.0, Frontend Architecture v1.0, Enterprise Security Architecture v1.0, Engineering Standards v1.0, Test Strategy v1.0 and ADR-111 DevOps Infrastructure. Every claim in this document is either traced to one or more source documents or explicitly marked as a RECOMMENDED addition where the sources are silent. Nothing is silently invented and nothing is silently dropped.

The verdict is direct: the PreOne product vision is coherent and strong, but the documentation set as it stands is NOT development-ready, because it contains two incompatible data-model generations, three conflicting technology-stack declarations, six different role taxonomies, four structurally different admission pipelines, and more than forty lower-grade contradictions in finance, attendance, communication and platform policies. Any team that starts building by reading these documents in sequence would build three different products. The purpose of this document is to resolve every conflict once, with reasoning, and to define the one flow that all teams implement.

1.1 Verdict by Layer

Table 1-1: Development-readiness verdict per architecture layer

Layer	State of the sources	Verdict	Blocking?
Product vision & terminology	Consistent across PRD/DDD/UI: AI-ready, multi-tenant Enterprise Preschool OS, 13-14 business domains	USABLE	No
Role model	Six incompatible taxonomies (PRD, BRC, UI, Security, ERD, Prisma)	MUST FIX	Yes
Tenant hierarchy	school=tenant vs tenant>school>branch vs tenant>branch	MUST FIX	Yes
Admission pipeline	4+ incompatible stage lists (PRD x2, DDD x2, UI steppers, ADR-037 workflow)	MUST FIX	Yes

Layer	State of the sources	Verdict	Blocking?
Student / parent lifecycle	Status lists differ across PRD, DDD, ERD, Prisma; no parent lifecycle anywhere	MUST FIX	Yes
Attendance flow	Status count conflicts (3 vs 5), edit model conflict (edit vs immutable+correction)	MUST FIX	Yes
Fees & finance	GST, late fee, refund approval, money precision all conflict; strongest rules exist in BRC	MUST FIX	Yes
Onboarding / setup / settings	Fragments exist (PRD FR-054, ADR-058 bootstrap, UI wizard) but no unified lifecycle	MUST FIX	Yes
AuthN / AuthZ	Login flows differ per doc; permission grammar has 4 competing formats	MUST FIX	Yes
Data model	Two generations: ERD+API (school_id, integer cents) vs Prisma+OpenAPI (tenantId, Decimal)	MUST FIX	Yes
API surface	~530 endpoints catalogued; envelope, paths and pagination differ between catalog and OpenAPI	MUST FIX	Yes
AI Center	Capabilities listed in PRD + 5 endpoints in API catalog; no persistence, no review workflow, ADR-148 missing	PARTIAL	Conditional
Growth Passport	Named in downstream materials but defined in ZERO source documents	UNDEFINED	Yes
Audit & security	Strongest area: Security Architecture + ADR-086 + BRC data rules agree on model and retention	USABLE	No

Layer	State of the sources	Verdict	Blocking?
DevOps / infra	ADR-111 complete and internally actionable; minor conflicts with Security doc (CSP, secrets store)	USABLE	No

1.2 The Ten Canonical Decisions That Fix the Product

Each decision below is fully derived and justified in the referenced section of this document. They are listed here so that a CTO can absorb the entire audit in one page.

Table 1-2: Canonical decisions at a glance

#	Decision	Where
D1	One application, one backend, RBAC-driven: platform console and school portal are route groups of the same system, gated by role. Parent portal is a mobile-first surface of the same backend.	Sec. 05
D2	Canonical hierarchy: Platform > Tenant (= School account) > Branch > Academic Year > Class/Section > Student. The extra school level in the API JWT spec is collapsed into Tenant; chains are modeled as multi-branch tenants.	Sec. 04
D3	Canonical 11 roles in 3 levels (Platform Admin; School Owner, Principal, Coordinator, Admission Counsellor, Teacher, Accountant, Receptionist, HR Manager, Transport Manager; Parent). All other role names in the sources map onto these.	Sec. 03
D4	One admission pipeline with five separated status domains: Lead Status, Application Status, Document Status, Approval Record, Enrollment Status. CRM funnel views are reporting projections, not statuses.	Sec. 10

#	Decision	Where
D5	Canonical tenant lifecycle: DRAFT > PENDING_ACTIVATION > ONBOARDING > ACTIVE > SUSPENDED > ARCHIVED, with optional TRIALING. Owner invitation is a user-level INVITED state, not a tenant state.	Sec. 06
D6	Attendance: 5 statuses (PRESENT, ABSENT, LATE, HALF_DAY, LEAVE); same- day edit with reason; after-day change only via approved CorrectionRequest; every row append-only + audited.	Sec. 15
D7	Money stored as integer paise everywhere; GST 0% on pure preschool tuition (BRC wins); late fee = BRC slab; refund approval = BRC matrix (Branch Admin <= INR 5,000; Branch Head <= INR 25,000; Director above), TAT 14 working days.	Sec. 18
D8	Permission grammar: module:action (lowercase, colon- separated), e.g. students:create, attendance:approve. All four competing grammars in the sources map to this one.	Sec. 24
D9	Growth Passport is canonically defined as the composition of Activities + Observations + Milestone Assessments + Achievements + Portfolio + Daily Timeline, rendered as a longitudinal parent-facing view and term PDF. Marked RECOMMENDED because no source defines it.	Sec. 17
D10	AI is a read-only intelligence layer over validated operational data: AI never writes core student, attendance or finance records; every AI output carries provenance (ai_draft) and requires human acceptance before it becomes operational data.	Sec. 20

1.3 What Each Team Should Read First

Table 1-3: Reading map per team

Team	Read first	Then
CTO / Tech Lead	Sec. 01, 02, 24, 31	Sec. 04, 05, 26, 27
Product Manager	Sec. 01, 02, 06-13, 31	Sec. 15-21
UI / UX	Sec. 23 (navigation + page specs)	Sec. 05, 10-21 for page-level behaviour
Backend	Sec. 24, 25, 26, 27	Sec. 10-22 for module logic
Frontend	Sec. 23, 05, 26	Sec. 10-21 for page behaviour
QA	Sec. 25, 28, 29	Sec. 24 (RBAC matrix)
DevOps	Sec. 32 (DevOps items)	ADR-111 (still valid), Sec. 22 audit pipeline

02. Current Documentation Problems

This section is the complete audit output. It classifies everything found in the source set into six buckets: what is already correct, what is inconsistent, what is duplicated, what is missing, what is ambiguous, and what needs a product decision. Every conflict is shown as Version A versus Version B (with sources) and the canonical ruling that the rest of this document implements. The same method applies throughout: the ruling always preserves the PreOne product vision and prefers the most specific, most recent, and most internally consistent source.

2.1 What Is Already Correct (Preserve As-Is)

- Product definition and positioning: AI-ready Enterprise Preschool Operating System, multi-tenant SaaS, 13-14 business domains, paperless and mobile-first objectives (PRD Sec. 2, DDD Sec. 4).
- Multi-tenant isolation principles: tenant_id discriminator on every table, PostgreSQL RLS as defense in depth, cross-tenant reads forbidden at the ORM layer, super-admin exemption audited (BRC R-PLT-001/008, ADR-043, Backend TD Sec. 21).
- Audit architecture: append-only audit log written in the same transaction as the business operation, hash chain, WORM cold storage, 7-year retention (Security Sec. 11, ADR-086, DDD AuditLog invariants).
- Token model: 15-minute RS256 access token, rotated single-use refresh token with family revocation on reuse, Redis revocation list (Security Sec. 6, API catalog, ADR-070/071).

- Soft delete: deleted_at column, no application-code hard deletes, retention job is the only deleter (ADR-047, Engineering Standards, Prisma middleware).
- Design system: Fluent-Metro synthesis, whitespace discipline, tile-based dashboards, card max 2 nesting levels, 150-250ms motion, WCAG 2.1 AA (UI Design Philosophy v1.0).
- Notification engine shape: domain events to outbox, Redis Streams fan-out, per-channel queues, at-least-once with idempotent consumers, DLQ after retries (Backend TD Sec. 14-15, ADR-027/034).
- Test strategy targets: every BRC business rule has at least one test, 10 critical E2E flows defined, RBAC allow/deny matrix encoded in test data (Test Strategy Sec. 11).

2.2 What Is Inconsistent (Conflict Register)

The register below lists every material conflict discovered between source documents, with the canonical ruling applied in this document. Conflicts marked CRITICAL would cause two teams to build incompatible systems.

Table 2-1: Master conflict register (Version A vs Version B vs Canonical)

#	Topic	Version A (source)	Version B (source)	CANONICAL RULING
C01	Backend stack	NestJS modular monolith, Prisma, BullMQ (Backend TD; DDD)	Java 21 / Spring Boot frozen 12 months (ADR-020, ADR Vol 1-2)	NestJS + Prisma + PostgreSQL 16 + Redis + BullMQ (Backend TD + Engineering Standards). ADR Vol 1-2 Java stack is a foreign template artifact; treat ADR-020 as superseded by the Backend TD stack freeze.
C02	Frontend stack	Next.js 16 App Router monorepo, apps/web (Frontend Architecture)	Next.js 16 frontend + API routes full-stack (PRD Sec. 2.5)	Next.js 16 frontend only; API always on the NestJS backend. PRD API-routes framing dropped.
C03	Tenant hierarchy	One school = one tenant; tenant owns branches (DDD, ERD)	tenant_id AND school_id as separate JWT claims (API catalog JWT)	Tenant = school account. JWT claims: tid, bid, ayid, uid. No separate school level. Chains = multi-branch tenants (Sec. 04).

#	Topic	Version A (source)	Version B (source)	CANONICAL RULING
C04	Role count / names	8 roles: Platform Admin, School Owner, Principal, Coordinator, Teacher, Accounts, Reception, Parent (PRD 7.1)	BRC enforces Admission Counsellor, Branch Head, Branch Admin, Director, Board, Finance Head, HR...; UI lists Super Admin, School Admin, Branch Admin, managers; Security lists Super Admin, Finance User, HR User; Prisma seeds PLATFORM_ADMIN, SCHOOL_ADMIN, ACCOUNTANT, RECEPTIONIST, LIBRARIAN	Canonical 11 roles (Sec. 03): PLATFORM_ADMIN; SCHOOL_OWNER, PRINCIPAL, COORDINATOR, COUNSELLOR, TEACHER, ACCOUNTANT, RECEPTIONIST, HR_MANAGER, TRANSPORT_MANAGER; PARENT. Mapping table provided. DIRECTOR/BOARD approval tiers become configurable approval levels, not roles, in v1.0.
C05	Admission pipeline	Lead > Enquiry > Counselling > Application > Documents > Approval > Fee > Student (DDD 20.2); Draft > Submitted > Verified > Approved/Rejected > Enrolled (DDD aggregate)	INQUIRY > APPLICATION > DOCUMENT_UPLOAD > PARENT_APPROVAL > BRANCH_APPROVAL > SCHOLARSHIP_APPROVAL > FEE_PAYMENT > ENROLLED (ADR-037); PRD CRM 6-stage and PRD Admissions 8-step; UI steppers	One pipeline, five separated status domains (Sec. 10). ADR-037 human-task timers (parent confirm 7d, internal approval 3d, escalation) are adopted as workflow SLAs. PRD Offer/Acceptance becomes Enrollment statuses OFFERED/ACCEPTED. Waitlist kept (DDD Approval entity + UI).
C06	Lead statuses	NEW > CONTACTED > QUALIFIED > CONVERTED / LOST (ERD)	Prisma adds NURTURE, APPLICATION_STARTED, DUPLICATE; OpenAPI shows New > Contacted > Interested > Visit Scheduled > ...	NEW, CONTACTED, QUALIFIED, NURTURE, CONVERTED, LOST, DUPLICATE (ERD base + Prisma additions). OpenAPI's Interested / Visit Scheduled are funnel activities tracked as lead activities, not statuses (Sec. 10.1).

#	Topic	Version A (source)	Version B (source)	CANONICAL RULING
C07	Student statuses	active / transferred / graduated / archived (PRD)	Provisional > Active > Promoted > Transferred > Archived (DDD); ACTIVE, INACTIVE, GRADUATED, TRANSFERRED, ARCHIVED (ERD); + SUSPENDED (Prisma)	PROVISIONAL > ACTIVE > INACTIVE > GRADUATED TRANSFERRED WITHDRAWN > ARCHIVED. Promotion is an EVENT (history entry), not a state. SUSPENDED folds into INACTIVE with reason (Sec. 12).
C08	Attendance statuses	present / absent / late / half-day / leave (PRD)	3 toggles present/absent/late (UI); + HOLIDAY, SUSPENDED (Prisma enum)	5 canonical statuses: PRESENT, ABSENT, LATE, HALF_DAY, LEAVE. HOLIDAY is a calendar concept, not a per-student status; Prisma extras dropped (Sec. 15).
C09	Attendance edit model	Edit allowed with reason; original preserved in audit (PRD AC-018)	Records immutable after submission; corrections via AdjustmentRecord (DDD)	Same-day: teacher edits in place with mandatory reason; next-day onward: CorrectionRequest with approval (attendance:approve) ; all corrections append-only + audited (Sec. 15.3).
C10	Money precision	Integer cents columns *_cents (ERD 4.2, all tables)	numeric(12,2) never integer paise (ERD 3.1 text); Prisma Decimal fields	Integer paise (bigint) everywhere. ERD 3.1 sentence is a doc bug contradicted by every ERD table; Prisma Decimal mapped to paise at migration. Display divides by 100 (Sec. 18.6).
C11	GST on fees	Item-wise CGST+SGST @ 18% (PRD R-FIN-002)	Pure preschool tuition GST-exempt 0%; transport/meal 5-18% per HSN (BRC R-FIN-007)	BRC wins: 0% on tuition/activity/exam for preschool; auxiliary services taxed per HSN as configured. PRD 18% flat dropped (Sec. 18.5).

#	Topic	Version A (source)	Version B (source)	CANONICAL RULING
C12	Late fee	Flat INR 100/day capped INR 1,000 or 5% (DDD)	Slabs: 1-7d 0; 8-30d INR 100; 31-60d INR 500; >60d INR 1,000 + service hold (BRC R-FIN-002)	BRC slab model, configurable per tenant via late_fee_rule (grace 7d default). DDD flat model dropped. >60d service hold requires Principal approval (Sec. 18.4).
C13	Refund approval	Principal <= INR 5k; Owner > INR 5k; TAT <= 7 days (PRD)	Branch Admin <= 5k; Branch Head <= 25k; Director > 25k; 14 working days (BRC R-APR-002); 2-tier > INR 5k + 90-day window (DDD)	BRC matrix mapped to canonical roles: Accountant initiates; PRINCIPAL approves <= INR 5,000; SCHOOL_OWNER approves <= INR 25,000; OWNER+finance committee above; TAT 14 working days; eligibility window 90 days (RECOMMENDED merge of DDD). (Sec. 18.4)
C14	Observation frequency	>= 3 per child per week (PRD BR-009)	>= 2 per student per week (DDD KPI); >= 1 per child per week with system alert (BRC R-ACD-004)	Hard minimum 1/week (BRC, system-enforced alert); operational target 2/week (DDD KPI on dashboards); 3/week aspirational only (Sec. 16.3).
C15	Observation photos	max 10 photos, 5 MB each (PRD)	max 5 attachments, 5 MB each (DDD)	5 attachments, 5 MB each (stricter, mobile-first). (Sec. 16.2)
C16	Report card approval	Teacher review > Principal approval, always (PRD, DDD)	Coordinator alone for Playgroup/Nursery; Coordinator+Principal for Jr/Sr KG (BRC R-ACD-010)	Coordinator quality review then Principal sign-off for ALL classes (PRD/DDD win; single invariant, no class-conditional logic). (Sec. 16.4)
C17	Fee reminder cadence	D-3, due date, D+7 (PRD)	D-7 Push, D-3 Push+SMS, overdue Push+SMS+Email+WhatsApp (BRC R-NOT-001)	BRC cadence (D-7, D-3, overdue+); add on-due-date reminder as RECOMMENDED. (Sec. 19.3)

#	Topic	Version A (source)	Version B (source)	CANONICAL RULING
C18	Message response SLA	<= 4 hours (PRD BR-002)	2 hours in school hours; next working day 10 AM otherwise (BRC R-COM-001)	BRC time-aware SLA wins. (Sec. 19.3)
C19	Response envelope	{ success, data, meta{page,pagesize, total,hasMore} } (API catalog)	Unwrapped resources + { data, meta{size,hasNext} } (OpenAPI); flat {success, errorCode, ...} (Backend TD)	Wrapped envelope everywhere: { success, data, error{code,message,field,details[]}, meta{page,pagesize, total,totalPages,hasMore}, traceId }. OpenAPI to be regenerated from this. (Sec. 26.2)
C20	Validation HTTP code	422 (Backend TD, Engineering Standards)	400 (Test Strategy API examples)	422 for DTO/semantic validation; 400 for malformed JSON; 409 conflicts; 403 permission. (Sec. 26.3)
C21	Permission grammar	students:create colon (Backend TD, ADR-005)	STUDENT_READ (Frontend); STUDENT.READ (Security); students.read (Eng Standards)	module:action lowercase colon (Sec. 24.2). Frontend consumes uppercase transform generated from the same source.
C22	Concurrent sessions	Max 3 (Frontend)	Max 5 (Security)	5 per user, oldest revoked (Security wins). (Sec. 05.5)
C23	MFA factors	SMS OTP rejected as insecure (ADR-072 decision)	SMS OTP 6-digit listed/enrolled (ADR-072 UI + Security Sec. 7)	TOTP mandatory for staff privileged roles; SMS OTP allowed only for Parent accounts; step-up MFA for high-risk operations. (Sec. 05.6)
C24	Password policy	Min 8, no composition rules, NIST (ADR-075 decision)	Min 12 + composition + history checks (ADR-075 UI; Security Sec. 5.2)	Min 12 with upper/lower/digit/symbol, last-5 reuse block, breach check; Argon2id. (Sec. 05.6)
C25	Invoice mutability	Issued invoices NEVER updated; corrections via CreditNote (Prisma)	PATCH /finance/invoices/{id} (draft only) (OpenAPI)	Immutable after ISSUE; PATCH allowed only while DRAFT; corrections via CreditNote/Refund. (Sec. 18.3)

#	Topic	Version A (source)	Version B (source)	CANONICAL RULING
C26	Bounded context count	13 (PRD) vs 8 (DDD) vs 12 (ADR-005) vs 14 (UI)	14 modules with API counts (Backend TD; Frontend)	Canonical 14 modules (Sec. 23.1, Sec. 27.1): identity, crm, admissions, student, academics, attendance, communication, finance, inventory, hr, administration, reports, settings, platform.
C27	Rule ID universe	DDD cites R-ADM-xxx, R-STD-xxx, R-DG-xxx, R-PLT-012, R-FIN-022 which do not exist in BRC; ~15 IDs reused with different meanings	BRC defines 176 rules across 12 categories	BRC v1.0 rule IDs are the only authoritative IDs. DDD citations to non-existent IDs are marked void; re-mapping table to be issued with BRC v1.1 (Sec. 31, OD-9).
C28	Seeded platform roles	PLATFORM_ADMIN, SCHOOL_ADMIN, PRINCIPAL, TEACHER, PARENT, ACCOUNTANT, RECEPTIONIST, LIBRARIAN (Prisma seeds)	Security/TS seed: Super Admin, School Owner, Finance User, HR User, Student...	Prisma seed set is the implementation baseline; extended per D3: add COORDINATOR, COUNSELLOR, HR_MANAGER, TRANSPORT_MANAGER; SCHOOL_ADMIN folds into SCHOOL_OWNER delegation; LIBRARIAN deferred to v1.1 (inventory module).
C29	PK generation	uuid_generate_v7 (ERD)	gen_random_uuid / uuid() v4 (Prisma, OpenAPI)	UUID v7 for new tables (time-sortable, index-friendly); existing v4 values kept. (Sec. 27.2)
C30	Audit log shape	user_id, action, module, entity, entity_id, old/new_values, ip, user_agent, request_id (ERD)	id, tenantId, entityType, entityId, action, oldValue, newValue, performedBy (Prisma); + role_snapshot, hash_chain (Security)	Superset record defined in Sec. 22.2 including role_snapshot, ip, user_agent, request_id, reason, hash_chain; Prisma model extended.

#	Topic	Version A (source)	Version B (source)	CANONICAL RULING
C31	AI persistence	5 AI endpoints (API catalog); feature flag ai.observations.enabled (ERD)	Zero AI tables in ERD/Prisma; ADR-148 AI Assistant referenced but absent from the doc set	New ai_insight + ai_usage tables with review states added to the canonical schema; ADR-148 to be authored (Sec. 20.4, Sec. 31).
C32	Domain framing	Preschool OS: Playgroup/Nursery/L KG/UKG, age 1.5-6 (PRD/BRC)	K-12 framing: grade-10, gradebook, transcripts, Class Grade 1-12 (ADR Vol 3-4, Prisma Class)	Preschool framing wins: programs Playgroup, Nursery, LKG, UKG, Daycare. Prisma Class model remapped to program/classroom semantics; K-12 vocabulary in ADR Vol 3-4 marked as template artifact. (Sec. 27.3)
C33	Number formats	Invoice INV-YYYY-NNNNN unique per school per FY (ERD); invoice INV-{BRANCH}-{FY}-{SEQ} (BRC)	student_code STU-YYYY-NNNNN (ERD) vs admissionNo ABC-0001 (Prisma)	Invoice: unique per school per FY (ERD wins). Receipt: unique per branch per FY (fee_receipt_counter). Student admission number: STU-YYYY-NNNNN unique per tenant. Application: APP-YYYY-NNNNN. (Sec. 18.3, Sec. 11)
C34	Webhook events	admission.approved, fee.paid, attendance.marked, invoice.generated, student.promoted (API catalog)	student.admitted, fee.paid, attendance.marked, invoice.generated, report-card published (OpenAPI)	Merged set of 7: admission.approved, student.admitted, fee.paid, invoice.generated, attendance.marked, student.promoted, reportcard.published. (Sec. 26.5)

2.3 What Is Duplicated

- Admission pipelines: defined 4+ times (PRD Marketing & CRM, PRD Admissions, DDD state machine, DDD aggregate invariant, ADR-037 workflow YAML, UI steppers). Canonical single pipeline in Sec. 10.
- Onboarding concepts appear as three unrelated things: PRD FR-054 tenant onboarding, UI M01 first-login wizard, UI M11 tenant creation stepper. Reunified in Sec. 06-09.

- Two communication data models: ERD announcement/chat_room family vs Prisma Message/Conversation family. One model ruled in Sec. 27.3.
- Two finance models: ERD fee_plan/fee_installment/invoice_items vs Prisma FeeStructure/Invoice/InvoiceLineItem. One model ruled in Sec. 27.3.
- Refund, late-fee and GST policies each defined 2-3 times across PRD, BRC and DDD with different numbers. Single rulings C11-C13.

2.4 What Is Missing

- Growth Passport: the term is used downstream but defined nowhere (no entity, no flow, no permission). Ruled in Sec. 17.
- Parent lifecycle: no source defines when a parent account is created, invited, activated or deactivated. Defined in Sec. 13.
- School/tenant status lifecycle: Prisma has PENDING_ACTIVATION, DDD has Trial>Active>Suspended>Offboarded>Archived; no document unifies them. Ruled in Sec. 06.
- Invitation and first-login password setup for staff and owners: absent everywhere (closest: StaffOnboarded event). Defined in Sec. 06.5 and Sec. 14.
- AI persistence and AI review workflow: endpoints exist, no tables, no states. Defined in Sec. 20.4.
- Account activation / suspension state machine for users: fragments only (ERD UserStatus, Security lockout). Ruled in Sec. 25.2.
- Transport: PRD defers to v1.1+, but DDD/UI/BRC ship full transport context and rules. v1.0 scope ruled in Sec. 31 (OD-4).

2.5 What Is Ambiguous

- Who can approve what: BRC uses Branch Head / Director / Board; PRD uses Principal / Owner. Mapped to canonical roles in Sec. 24.4.
- Whether Coordinator is a distinct role or a Principal duty (PRD has Coordinator; Security/TS seed does not). Ruled as distinct optional role, Sec. 03.3.
- Whether DELETE on students hard-deletes: API catalog offers DELETE /students (soft-delete 7y DPDP); UI shows Archive; DDD forbids app hard deletes. Ruled in Sec. 12.4.
- Whether invoice numbering is per school or per branch (C33). Ruled per school; receipts per branch.
- Whether the parent app is a separate mobile app or responsive web: FA scopes React Native as future; PRD calls parent app phone-first. Ruled in Sec. 05.2 (D1).

2.6 What Needs A Product Decision

Eleven decisions cannot be resolved from the sources alone because they are genuine product choices, not inconsistencies. They are registered with options and recommendations in Sec. 31 (Open Decisions) and summarized here: transport in/out of v1.0; observation frequency target; refund eligibility window; invoice numbering scope; Growth Passport v1.0 composition sign-off; AI vendor set (Z.ai GLM primary + OpenAI reserve vs adding Anthropic); MFA factor set for parents; role naming for the finance function (ACCOUNTANT vs Finance User); GST configuration surface per tenant; custom roles per tenant in v1.0 or v1.1; and the BRC rule-ID reissue.

03. Corrected Role Architecture

The sources disagree on how many roles PreOne has and what they are called. The PRD declares eight primary roles; the BRC operationally enforces at least fifteen more; the UI document grants modules to Super Admin, School Admin, Branch Admin and various managers; the Security Architecture lists eight different ones; the ERD user_type has six; the Prisma schema seeds eight different codes. This section ends that drift. The canonical model has exactly eleven roles in three levels, and every role name found anywhere in the sources maps onto one of them. Role, Permission, User, Tenant and Branch are distinct concepts; their separation is defined in Sec. 24.1.

3.1 Canonical Role Hierarchy

Table 3-1: Canonical roles (11)

Level	Role code	Role name	Purpose	Primary surface
PLATFORM	PLATFORM_ADMIN	Platform Admin	Operates PreOne SaaS: creates tenants, activates schools, manages subscriptions, platform flags, cross-tenant audit	Platform console
SCHOOL	SCHOOL_OWNER	School Owner	Business owner of the school tenant; owns settings, approvals above Principal tier, finance oversight	Admin portal (desktop)

Level	Role code	Role name	Purpose	Primary surface
SCHOOL	PRINCIPAL	Principal	Academic + operational head: admissions approval, attendance oversight, staff coordination, communication approvals	Admin portal + tablet
SCHOOL	COORDINATOR	Academic Coordinator	Multi-class academic supervision: lesson plan review, observation quality review, class-level broadcasts	Admin portal + tablet
SCHOOL	COUNSELLOR	Admission Counsellor	Owns CRM leads and admission applications up to verification; cannot grant final admission approval	Admin portal
SCHOOL	TEACHER	Teacher	Classroom caregiver: attendance marking, activities, observations, daily reports, class communication	Tablet-first
SCHOOL	ACCOUNTANT	Accountant	Fee plans, invoices, payments, receipts, refunds initiation, finance reports	Admin portal (desktop)
SCHOOL	RECEPTIONIST	Receptionist	Front desk: walk-in lead capture, visitor log, gate pass, enquiry intake	Admin portal
SCHOOL	HR_MANAGER	HR Manager	Staff records, payroll runs, leave approval, staff documents	Admin portal
SCHOOL	TRANSPORT_MANAGER	Transport Manager	Routes, vehicles, drivers, transport attendance (v1.1 feature-flagged)	Admin portal

Level	Role code	Role name	Purpose	Primary surface
PARENT	PARENT	Parent	Child timeline, fees view + payment, chat with school, announcements, consent management	Parent app (phone-first)

3.2 Mapping Of Every Source Role Name

Every role name found in any source document maps to exactly one canonical role. Names that appear in operational rules but not in role tables (Branch Head, Branch Admin, Director, Board, Finance Head, Center Head) are mapped below; where the mapping is a product simplification it is flagged.

Table 3-2: Source role name to canonical role mapping

Source name (doc)	Canonical role	Note
Super Admin (UI 3.1, Security 8.2)	PLATFORM_ADMIN	Name unified; two names for one concept in sources
Platform Admin (PRD 7.1)	PLATFORM_ADMIN	—
School Owner (PRD)	SCHOOL_OWNER	—
Owner (BRC review list)	SCHOOL_OWNER	—
Branch Head (BRC)	PRINCIPAL	Single-campus equivalent; multi-branch chain keeps Principal per branch
School Admin (UI, Prisma SCHOOL_ADMIN)	SCHOOL_OWNER	Delegated admin privileges via role bundle, not a separate role
Coordinator (PRD), Academic Coordinator (UI)	COORDINATOR	Optional role; small schools may leave unassigned (duties fall to Principal)
Admission Counsellor (BRC), Sales (UI M02)	COUNSELLOR	—
Accounts / Accounts Team (PRD, BRC), Accountant (DDD, Prisma)	ACCOUNTANT	Finance Head (BRC) maps to SCHOOL_OWNER finance approvals in v1.0
Reception (PRD, BRC)	RECEPTIONIST	—
HR (BRC), HR Manager (UI), HR User (Security/TS)	HR_MANAGER	—
Transport Manager (UI), Transport In-charge (BRC)	TRANSPORT_MANAGER	Feature-flagged v1.1
Parent (all docs)	PARENT	—

Source name (doc)	Canonical role	Note
Student (Security 8.2 list)	not a login role in v1.0	Students are data subjects; student portal deferred (RECOMMENDED)
Director / Board (BRC approvals)	approval tiers, not roles	Modeled as approval level 3/4 in approval workflows (Sec. 10.4)
Center Head (DDD role examples)	PRINCIPAL	One-off naming drift in DDD
Kitchen In-charge / Nap Supervisor / Health Attendant / Security / IT / Care Staff (BRC)	operational duty tags, not v1.0 roles	Handled as staff designations with task checklists, not RBAC roles
TEACHER/ADMIN/SUPPORT legacy hard-coded (ADR-064)	replaced by canonical set	ADR-064 itself mandates the system+custom catalog that makes this possible

3.3 Per-Role Authority Profile

The table below defines, for every canonical role, the attributes requested by the audit brief: who creates the account, who deactivates it, where it logs in, and its authority across the ten permission families. The full action-level matrix is in Sec. 24.3; this table is the role-level contract.

Table 3-3: Role authority profile (C=create, R=read, U=update, D=delete-soft, A=approve, F=financial ops, S=student ops, M=messaging, RP=reports, AI=AI use)

Role	Created by	Deactivated by	Login area	Permissions summary
PLATFORM_ADMIN	Seeded by platform bootstrap	Another PLATFORM_ADMIN	Platform console	All modules cross-tenant (audited); no routine student data access; tenant lifecycle full; platform flags full; audit visibility full incl. security events
SCHOOL_OWNER	PLATFORM_ADMIN invites	PLATFORM_ADMIN / self	Admin portal	C/R/U on school settings, staff, fee structures; A on refunds > Principal tier and discounts > 15%; F full oversight; S full; M broadcast to all; RP full; AI full; audit full (school scope)

Role	Created by	Deactivated by	Login area	Permissions summary
PRINCIPAL	SCHOOL_OWNER invites	SCHOOL_OWNER	Admin portal + tablet	C/R/U on students, classes, timetables; A on admissions, corrections, incidents; F read + refund initiate; S full; M broadcasts + announcements approval; RP school scope; AI full; audit read
COORDINATOR	SCHOOL_OWNER / PRINCIPAL	SCHOOL_OWNER / PRINCIPAL	Admin portal + tablet	R/U academics (lesson plans, timetables); A on observation quality flags; S R/U in assigned classes; M class-level broadcast; RP academic scope; AI use (draft + review)
COUNSELLOR	SCHOOL_OWNER / PRINCIPAL	SCHOOL_OWNER / PRINCIPAL	Admin portal	C/R/U on leads, enquiries, applications, documents; U application until SUBMITTED lock; A none (verify only); F none; S create-on-convert only; M lead follow-ups; RP funnel scope; AI lead-score assist
TEACHER	SCHOOL_OWNER / PRINCIPAL	SCHOOL_OWNER / PRINCIPAL / HR_MANAGER	Tablet-first	C/R/U attendance (own class, same-day), activities, observations, daily reports; A none; F none; S read own class; M class parents chat; RP own class; AI draft tools only

Role	Created by	Deactivated by	Login area	Permissions summary
ACCOUNTANT	SCHOOL_OWNER	SCHOOL_OWNER	Admin portal	C/R/U fee plans, invoices, payments, receipts; A refunds <= tier 1 initiation; F create invoices/payments/refund requests; S read fee-relevant student fields; M fee reminders (system-templated); RP finance scope; AI none on PII
RECEPTIONIST	SCHOOL_OWNER / PRINCIPAL	SCHOOL_OWNER / PRINCIPAL	Admin portal	C leads (walk-in), visitors, gate passes; R student directory (contact fields); A none; F receipt reprint only; S read contact; M none beyond templates; RP visitor scope; AI none
HR_MANAGER	SCHOOL_OWNER	SCHOOL_OWNER	Admin portal	C/R/U staff, payroll runs, leaves; A leave requests <= tier; F payroll execute (Owner countersign); S none; M staff announcements; RP HR scope; AI none
TRANSPORT_MANAGER	SCHOOL_OWNER	SCHOOL_OWNER	Admin portal	C/R/U routes, vehicles, drivers, route-student mapping; R transport attendance; M transport delay alerts (triggered); RP transport scope; AI none (v1.0)
PARENT	System, at admission approval	PRINCIPAL / PLATFORM_ADMIN	Parent app	R own children only (timeline, fees, attendance, announcements); U own profile + communication preferences; C leave requests, fee payments (own), consents; M chat with own child's teachers; AI none

CANONICAL RULE No school-level role can create, deactivate or delete a PLATFORM_ADMIN, and no PLATFORM_ADMIN performs routine operational data entry

inside a tenant (their writes are limited to tenant lifecycle and configuration, all audited). Custom roles per tenant are RECOMMENDED for v1.1 using the system+custom catalog of ADR-064, with the no-escalation guard: a custom role can never grant a permission absent from the system catalog.

04. Corrected Tenant Architecture

PreOne is a multi-tenant SaaS. The sources agree that tenant isolation is non-negotiable (ADR-001 principle 1) but disagree on the levels between platform and student. The API document assumes three levels (tenant, school, branch); the ERD anchors everything on school; the DDD says one school equals one tenant. This section fixes the canonical hierarchy and assigns every data category to exactly one ownership level, so that scoping, RLS policies and UI scope-switchers can be generated mechanically.

4.1 Canonical Hierarchy

```

PREONE PLATFORM
| platform-level: subscription plan catalog, global config, feature flags,
tenant registry
v
SCHOOL TENANT (Tenant = the school's account; 1 tenant = 1 school business)
| tenant-level: branding, settings, staff/users, roles, fee structures,
subscription, theme
v
BRANCH (physical campus; chains have many, single-campus schools have exactly
one)
| branch-level: timings, classrooms, local inventory, visitors, branch
announcements
v
ACADEMIC YEAR (tenant-level entity; partitions attendance, invoices,
promotions)
| year-level: sessions/terms, calendars, fee cycles, report cards
v
CLASS / SECTION (branch + academic-year scoped)
| class-level: timetables, attendance sheets, observation groups
v
STUDENT (branch + class + academic-year scoped; globally unique identity)
| student-level: profile, guardians, documents, observations, growth
passport, invoices

```

Two consequences follow. First, the separate school level that appears in the API catalog JWT (tenant_id plus school_id) is collapsed: the tenant IS the school account. Second, a chain is not a special object; it is one tenant with many branches, which is exactly what the ERD and Prisma already model, and Prisma's optional parentTenantId self-reference is retained only to support future franchise groupings for reporting, never for data inheritance.

4.2 Where Everything Belongs

Table 4-1: Data ownership by level

Data category	Owning level	Rationale
Subscription, plan, billing to PreOne	Platform > Tenant record	Platform bills the tenant, never the branch
Branding, theme, notification prefs	Tenant (branch may override subset)	UI theme cascade: platform > subscription > school > branch > user
Users (staff) and role assignments	Tenant; assignment scope = TENANT, BRANCH or CLASS	One person can serve two branches of the same tenant with two scoped assignments
Parents / guardians	Tenant	A guardian can be linked to multiple students across branches of the same tenant
Fee structures / fee plans	Tenant (branch may scope applicability)	ERD fee_plan.branch_id NULL = school-wide
Academic years / sessions / calendars	Tenant	Promotion, fee cycles and attendance partitioning all key on it
Programs, classes, sections	Branch + academic year	A class exists at a campus, in a year
Students	Branch (current), tenant (identity)	current_branch_id on student; history preserves every branch
Attendance, observations, daily logs	Branch + class + academic year	High-volume, partitioned by academic year (ADR-046)
Invoices, payments, receipts	Tenant identity, branch counter for receipts	Invoice number per school per FY; receipt counter per branch per FY
Transport	Branch (v1.1 feature-flagged)	Routes start at a campus
Documents (student/staff)	Tenant (storage prefix tenantId/studentId)	S3 keys partitioned by tenant per Security Sec. 10
Communication (chats, broadcasts)	Tenant; audience scoped to branch or class	Broadcast audience enum already models ALL_PARENTS / BRANCH_PARENTS / CLASS_PARENTS
AI insights	Tenant, derived	AI reads tenant-scoped data only; outputs are tenant rows
Audit logs	Tenant, mirrored to platform SIEM	Immutable, hash-chained, 7-year retention

4.3 Multi-Tenant Isolation Contract

Isolation is enforced in depth, and every layer is independently testable. The contract below merges BRC R-PLT-001/006/008, ADR-043/044/045 and the Backend TD Sec. 21 into one

normative list; where sources differed on layer names, the union is taken because they are complementary, not contradictory.

- Layer 1 - JWT: every token carries tid (tenant), bid (active branch), ayid (active academic year), uid, roles[], permissionsVersion. No query may run outside the token scope.
- Layer 2 - Request guard: PermissionsGuard resolves effective permissions from cache (60-second freshness budget) and rejects 403 before any data access; TENANT_MISMATCH on header/claim conflict returns 403, and cross-tenant probes return 404 to avoid existence leaks.
- Layer 3 - ORM middleware: Prisma middleware auto-injects tenantId on every query and blocks unscoped queries; soft-delete filter (deletedAt null) auto-applied.
- Layer 4 - PostgreSQL RLS: row-level security policies compare tenant_id against current_setting(app.tenant_id); bypass role exists only for the reporting service account and is logged via event trigger.
- Layer 5 - Storage and cache: S3 keys prefixed tenants/{tenantId}/...; Redis keys prefixed tenant_id; search indices per tenant; theme and settings caches keyed per tenant.
- Exception path: PLATFORM_ADMIN cross-tenant access requires explicit impersonation with reason, renders a persistent Viewing-as-Tenant banner, and writes an audit record (BRC R-PLT-001 exception clause).

CANONICAL RULE A user authenticated under tenant A can never read or write tenant B data through any API, report, export, webhook payload or AI insight. Platform-level analytics run only on anonymized aggregates ($k \geq 10$) per BRC R-DAT-012.

05. Authentication Architecture

The sources describe the same authentication machinery three times with different details: the Backend TD defines an email+password flow, the Security Architecture adds phone+OTP and MFA before token mint, and the Frontend Architecture describes a post-login sequence that never mentions tenant or role resolution. This section merges them into one architecture and answers the audit brief question directly: PreOne uses ONE application with RBAC-driven experiences (Option B), not separate platform and school applications, and not a loose multi-portal federation.

5.1 Architecture Options Evaluated

Table 5-1: Application architecture options

Option	Description	Evidence for	Evidence against	Ruling
A: Separate apps	Distinct platform-admin app and school-admin app, shared DB	Cleaner blast radius for platform ops	Doubles routes, builds, deployments, RBAC drift; sources never define two admin apps; FA monorepo has single apps/web with a platform module	REJECTED
B: One app + RBAC	Single Next.js app; /platform/* and /app/* route groups; guards resolve experience from role	FA route map already contains platform module; middleware permission gate exists; single deployment, single session model; matches the implemented system	Requires strict route-guard tests (provided in Test Strategy)	CANONICAL
C: Multiple portals	Separate portals (admin, teacher, parent) with shared backend	PRD names teacher/parent surfaces; mobile-first	Three frontends for one permission model multiplies UI work; parent surface is a view subset of the same API, not a different domain	PARTIAL: same backend, parent app is a phone-first client of the same API (v1.1 native), teacher tablet is the same responsive app

CANONICAL RULE (D1) One Next.js 16 application, one NestJS backend. Role determines route group: PLATFORM_ADMIN lands on the platform console; every school role lands on the admin portal; PARENT lands on the parent surface. No school role can reach /platform/* and PLATFORM_ADMIN has no school dashboard.

5.2 Canonical Login Flow (Every Branch)

The flow below is normative. Each decision node names the state that produces each outcome; corresponding error codes are in Sec. 26.3.

```

LOGIN REQUEST (email+password, or phone+OTP for parents)
  v
[1] CREDENTIAL CHECK -- fail x5 --> account LOCKED (15 min; escalate per
Security 5.2) --> END
  v pass
[2] RESOLVE USER STATUS

```

```

    PENDING_VERIFICATION -----> show invitation-accept / set-password
screen --> END
    INACTIVE / DELETED -----> ACCOUNT_DISABLED error --> END
    LOCKED -----> LOCKED error + unlock path --> END
    ACTIVE -----+
v
[3] RESOLVE TENANT
    user has tenant membership -> load tenant --> [4]
    user has NO membership and PLATFORM_ADMIN flag -> tenant=null -> [4]
    tenant SUSPENDED -----> SCHOOL_SUSPENDED screen (read-only) --> END
    tenant PENDING_ACTIVATION -> TENANT_PENDING screen (owner sees activation
checklist) --> END
v
[4] RESOLVE ROLE + BRANCH
    multi-role user -> role picker (remembers last) ; default = highest
precedence
    multi-branch user -> branch picker; sets bid claim
v
[5] LOAD PERMISSIONS
    effective permission set resolved (role bundles + scoped assignments),
cached 60s,
    permissionsVersion stamped into JWT
v
[6] MFA (risk-based)
    privileged roles (PLATFORM_ADMIN, SCHOOL_OWNER, PRINCIPAL, ACCOUNTANT,
HR_MANAGER) -> TOTP challenge
    parent logins -> optional SMS OTP; step-up on anomaly
v
[7] SESSION + TOKENS (access 15 min RS256; refresh 30d rotated single-use,
HttpOnly cookie)
v
[8] ROUTE BY ROLE
    PLATFORM_ADMIN -----> /platform (tenant registry, onboarding
console, platform audit)
    school roles -----> /app/dashboard (or /app/onboarding if
tenant.onboarding_completed=false)
    PARENT -----> /parent/home
    onboarding_completed=false AND role=SCHOOL_OWNER -> /app/onboarding
wizard

```

5.3 First-Login, Invitation And Password Setup

No source document defines how a newly created staff user receives credentials; this is a mandatory gap fill. The canonical mechanism is invitation-based and identical for owners and staff, with role-dependent expiry.

- Creation: an authorized creator (PLATFORM_ADMIN for owners; SCHOOL_OWNER/PRINCIPAL for staff) enters name, contact and role; the

system creates the user in PENDING_VERIFICATION and sends a signed invitation link.

- Invitation link: single-use, expires in 72 hours (owner) or 7 days (staff); resend allowed by the creator; link binds to the email/phone it was issued to.
- First login: user sets password (policy in 5.6), completes profile, and is walked into the role-appropriate tour; owners additionally enter the school onboarding wizard (Sec. 07).
- Activation: the user becomes ACTIVE on first successful password set. Deactivation and re-activation are separate admin actions (Sec. 25.2).
- Password reset: request issues a 15-minute single-use signed token; success revokes all sessions (family-wide) and notifies email+SMS for privileged roles.

5.4 Session Handling And Logout

- Access token 15 minutes, RS256, claims sub, tid, bid, ayid, roles[], permissionsVersion, sid.
- Refresh token 30 days, single-use rotation; reuse of a rotated token revokes the entire refresh family (family_id) and forces re-login everywhere.
- Concurrent sessions: max 5 per user; a sixth login revokes the oldest (Security wins over FA's 3).
- Idle timeout 30 minutes with a soft warning at 25; absolute session cap 8 hours.
- Logout blacklists both tokens and deletes the Redis session record; admin force-logout bumps permissionsVersion and revokes sessions server-side.

5.5 Unauthorized, Suspended And Inactive Behaviours

Table 5-2: Access-state behaviours

Condition	API behaviour	UI behaviour
No/invalid token	401 TOKEN_MISSING / TOKEN_INVALID	Redirect /login with return URL
Token expired	401 TOKEN_EXPIRED (refreshable)	Silent refresh; on failure redirect /login
Permission missing	403 PERMISSION_DENIED	403 page explaining why + request-access link; nav item hidden
Branch scope violation	403 SCOPE_VIOLATION	403 page; branch switcher offered if user holds another branch
Cross-tenant probe	404 (existence hidden)	Not-found page

Condition	API behaviour	UI behaviour
User INACTIVE/DELETED	401 ACCOUNT_DISABLED	Blocked at login; sessions already revoked
User LOCKED	423 LOCKED with retry-after	Lock screen with unlock instructions
Tenant SUSPENDED	Read endpoints 200 (read-only), writes 403 TENANT_SUSPENDED	Persistent read-only banner; billing contact card for owner
Tenant PENDING_ACTIVATION	Config APIs allowed; operational APIs 403 TENANT_PENDING	Activation checklist for owner; console view for platform admin

5.6 Credential Policies

- Password: minimum 12 characters with upper, lower, digit and symbol; last 5 reuse blocked; breach-corpus check; hashed with Argon2id (m=64MB, t=3) with bcrypt cost-12 fallback. This resolves C24 in favour of the stricter source; ADR-075's 8-char NIST variant is rejected for a child-data platform.
- MFA: TOTP mandatory for the five privileged staff roles; SMS OTP permitted only for PARENT accounts; step-up MFA (fresh 5-minute verification) required for high-risk operations: bank-detail change, bulk data export, refund execution, role assignment. This resolves C23.
- Device binding: first login from a new device records a fingerprint; unrecognized device triggers OTP step-up even with valid credentials.

06. School Creation Flow

School creation is the platform's core SaaS motion: a PLATFORM_ADMIN (or a self-serve signup, feature-flagged) creates a tenant, provisions the primary branch, invites the owner, and the school then onboards itself. The sources supply all the pieces (PRD FR-054 workflow, ADR-058 bootstrap endpoint, UI M11 wizard, Prisma tenant statuses, DDD lifecycle) but never assemble them. This section is the canonical assembly.

6.1 Canonical Lifecycle And Statuses

Table 6-1: Tenant status model (canonical)

Status	Meaning	Entered by	Allowed exits
DRAFT	Being created in the console form; not yet committed	Platform admin opening create form	PENDING_ACTIVATION (commit), discarded (purge)
PENDING_ACTIVATION	Committed; awaiting KYC/subscription activation	Create committed	TRIALING (trial path), ONBOARDING (activated)

Status	Meaning	Entered by	Allowed exits
TRIALING (optional)	14-day trial, limited features, watermarked	Trial signup path (BRC R-PLT-009 exception)	ONBOARDING (converted), ARCHIVED (expired)
ONBOARDING	Active subscription; owner completing guided setup	Activation or conversion	ACTIVE (checklist complete)
ACTIVE	Live school; daily operations running	Onboarding checklist complete	SUSPENDED, ARCHIVED
SUSPENDED	Writes blocked (payment failure or violation)	Platform admin action or automated dunning	ACTIVE (resolved), ARCHIVED (after 30-day window)
ARCHIVED	Offboarded; data export delivered; retention clock running	Offboarding flow (BRC R-PLT-010)	Hard purge after retention (irreversible)

Statuses are drawn from the sources with two normalizations. PENDING_ACTIVATION is taken from the Prisma tenant enum because it is the only source that names the pre-activation state; DDD's Offboarded is renamed ARCHIVED to match the student-lifecycle vocabulary; INVITED is deliberately NOT a tenant status (it is the owner-user's PENDING_VERIFICATION state) and SETUP_IN_PROGRESS is not adopted because ONBOARDING already covers it, per the brief's instruction to keep only justified states. Tenant user states and subscription states remain separate machines (Sec. 25).

6.2 Creation Steps

```

PLATFORM ADMIN > Create School
  v
STEP 1 School details (name, contact, city, programs offered)
[validate: unique name/email]
  v
STEP 2 Plan selection (Free/Starter/Pro/Enterprise/White-Label)
[validate: plan active]
  v
STEP 3 Primary branch (name, code, address, timings)
[validate: branch code unique per tenant]
  v
STEP 4 Owner identity (name, email, phone)
[validate: email not already a user]
  v
STEP 5 Review > COMMIT
      POST /api/v1/platform/tenants (Idempotency-Key header mandatory)
  v
TENANT = DRAFT -> PENDING_ACTIVATION
  * tenant row + primary branch row + default role catalog + permission bundles
  seeded (ADR-058)
  * owner user created PENDING_VERIFICATION; invitation email/SMS sent
  * audit: TENANT_CREATED (actor, payload hash)

```

```

v
ACTIVATION_GATE (BRC R-PLT-009)
  business proof = true AND signatory verified = true AND payment method
  valid = true
    yes -> ONBOARDING (owner first login opens wizard) [trial path ->
  TRIALING]
    no -> stays PENDING_ACTIVATION (checklist shown to platform admin)

```

6.3 Activation, Suspension And Offboarding

- Suspension triggers: subscription overdue past 7-day grace (automated), compliance violation, or explicit platform action; all writes across the tenant return 403 TENANT_SUSPENDED; reads remain available for data export.
- Resumption: on payment success or violation clearance, tenant returns to ACTIVE; a full audit record links suspension and resumption with reasons.
- Offboarding: 30-day window with data export (students, invoices, observations, documents); then tenant = ARCHIVED; automated purge completes per DPDP retention; purge is irreversible and preceded by a mandatory backup (DDD tenant invariants).

07. Onboarding Flow

Onboarding is the owner's first-time guided configuration, entered automatically after the first owner login when the tenant is in ONBOARDING. Its purpose is narrow: get a school from zero to operationally live with the minimum viable configuration, using sensible defaults so nothing is blocking. Onboarding never duplicates day-to-day configuration screens; it is a wizard over the same APIs the Setup pages use (Sec. 08-09 keep the separation crisp).

Table 7-1: Onboarding wizard steps

#	Step	What is configured	Default behaviour	Skippable?
1	Welcome & profile	Owner name display, school contact, logo upload	Logo optional; placeholder branding applied	No
2	Academic year	Current academic year + term dates	Auto-created from current date (April-March default)	No
3	Programs & classes	Enable programs (Playgroup...UKG/Daycare); create classes with capacity	Standard program set pre-checked with default age bands	No

#	Step	What is configured	Default behaviour	Skippable?
4	Fee structure	First fee plan per program (heads, amount, frequency)	Blank starter plan offered; can be assigned later	Yes (flag set)
5	Staff invites	Invite Principal first; others optional	Minimum one Principal before first admission approval	Partial (Principal required to go live)
6	Roles confirmation	Confirm bundle mapping for invited staff	Canonical defaults (Sec. 24)	No
7	Communication setup	Sender IDs, templates, quiet hours	System defaults; DLT templates linked where applicable	Yes
8	Consent & compliance	DPDP consent text, media consent, pickup authorization policy	Platform-default consent text versioned	No
9	Review & go live	Summary; GO LIVE action	Sets tenant.onboarding_completed=true; status ACTIVE	No

Exit criteria for ONBOARDING: academic year exists, at least one class exists, at least one Principal is ACTIVE, and consent text accepted. Fee structure and communication setup may remain incomplete; the dashboard shows a persistent setup-progress card until they are done. Re-entering the wizard later is possible from Settings > Onboarding progress, which makes the wizard a resumable checklist rather than a one-shot gate.

08. Setup Flow

Setup is the standing administrative configuration of the school: the pages a Principal or Owner returns to whenever the structure of the school changes. Setup is operational, not cosmetic: every item here affects runtime behaviour of other modules. The list below is derived from the PRD Settings domain description, the UI M13 categories and the ERD settings tables, deduplicated against onboarding (which calls the same APIs but is wizard-shaped).

Table 8-1: Setup responsibilities

Setup area	Managed by	Affects	Notes
Academic years & terms	PRINCIPAL / SCHOOL_OWNER	Attendance partitioning, fee cycles, promotion, report cards	Exactly one ACTIVE year; future year can be prepared
Branches	SCHOOL_OWNER	Scope switcher, class trees, branch receipts counter	Multi-branch only on plans that include it
Programs, classes, sections	PRINCIPAL / COORDINATOR	Admission eligibility, classroom assignment, attendance sheets	Age bands validated against BRC eligibility rules
Fee heads, plans, installments	ACCOUNTANT / SCHOOL_OWNER	Invoicing, reminders, discounts	Changes versioned; existing invoices unaffected
Discount & scholarship rules	SCHOOL_OWNER (approval tiers per Sec. 18.4)	Fee calculations	BRC rule set configurable per tenant
Staff & role assignments	SCHOOL_OWNER / HR_MANAGER	RBAC, attendance marking rights, approvals	Scoped assignments (tenant/branch/class)
Approval workflow config	SCHOOL_OWNER	Refund, discount, waiver, announcement approval tiers	Levels map to canonical roles (Sec. 10.4)
Integrations	SCHOOL_OWNER (PLATFORM_ADMIN for platform keys)	SMS/WhatsApp/email providers, payment gateway	Per-tenant provider routing (ADR-009)
Custom attributes / groups	PRINCIPAL	Student/staff extensible fields	JSONB-backed, form-builder UI (ADR-052)

09. Settings Flow

Settings is the ongoing preferences layer: things that change how the product looks and notifies, not what the school's structure is. Keeping Settings, Setup and Onboarding separate prevents the duplicate-page problem the audit brief warns about; the responsibility matrix below is the normative boundary, and any new configuration feature must be placed in exactly one of the three.

Table 9-1: Onboarding vs Setup vs Settings responsibility matrix

Concern	Onboarding (first-time wizard)	Setup (administrative config)	Settings (preferences)
School profile & branding	Logo + name (step 1)	-	Branding, theme, favicon, login background

Concern	Onboarding (first-time wizard)	Setup (administrative config)	Settings (preferences)
Academic year	Create current year	Manage years/terms/holidays	-
Branches	Primary branch already exists (created at tenant creation)	Add/edit branches	-
Classes	Create initial classes	Manage classes/sections all year	-
Fee structure	First plan optional	Full fee management	-
Staff	Invite Principal (minimum)	Staff CRUD, role bundles, scopes	-
Roles & permissions	Confirm defaults	Adjust assignments	Custom role editor (v1.1)
Notifications	Templates & sender setup	-	Quiet hours, per-user preferences, channel toggles
Consents & compliance	Consent text acceptance	Compliance documents & reminders	Consent text versioning
Integrations	-	Provider credentials & routing	Personal API tokens (admin)
Billing (PreOne subscription)	-	-	Plan, invoices, payment method (Owner + Platform)
Security	-	-	MFA enrollment, sessions, trusted devices, password
Onboarding progress	IS the wizard	-	Progress card + re-entry point

CANONICAL RULE (RULE 15) A configuration screen belongs to exactly one of the three surfaces. Onboarding embeds Setup APIs but never re-implements forms; Settings never changes structural data (that is Setup); Setup never duplicates preference toggles (those are Settings).

10. Admissions Flow

Admissions is the most conflicted area of the documentation set: the PRD defines two pipelines (Marketing & CRM and Admissions), the DDD defines two more (a workflow state machine and an aggregate status list), the UI adds two steppers, the BRC adds sub-status flags, and ADR-037 contributes a workflow-engine pipeline with human-task timers. The canonical ruling keeps ONE pipeline but formally separates the five status domains the brief demands, so that CRM funnel reporting and admission processing never share a vocabulary again.

10.1 The Five Status Domains

Table 10-1: Lead Status (CRM domain; owned by counsellor)

Status	Meaning	Next states
NEW	Captured (form, walk-in, campaign, referral); not yet contacted	CONTACTED, DUPLICATE (auto-merge candidate), LOST
CONTACTED	First outreach done; activity logged	QUALIFIED, NURTURE, LOST
QUALIFIED	Interest + basic fit confirmed; visit scheduled as an ACTIVITY	CONVERTED (application created), NURTURE, LOST
NURTURE	Long-cycle prospect in drip follow-up	QUALIFIED, LOST
CONVERTED	Application created from lead (terminal; lead retained for reporting)	-
LOST	Closed with mandatory lost_reason	-
DUPLICATE	Merged into another lead (phone+name match per BRC R-ELG-003)	-

Table 10-2: Application Status (admission processing domain)

Status	Meaning	Next states
DRAFT	Created (by counsellor from lead, or public form), not submitted	SUBMITTED, CANCELLED
SUBMITTED	Form locked (BRC R-ADM-006 analogue); fees/receipt as configured	DOCUMENT_PENDING
DOCUMENT_PENDING	Checklist incomplete (birth cert, photos, medical, parent ID)	DOCUMENT_VERIFIED, REJECTED (docs never provided in window)
DOCUMENT_VERIFIED	All checklist items verified (PENDING>VERIFIED per document)	UNDER_REVIEW
UNDER_REVIEW	In counselling + approval workflow	APPROVED, REJECTED, WAITLISTED
APPROVED	Approval chain complete; offer generated	CONVERTED (enrollment), CANCELLED (offer expired)
REJECTED	Rejected with reason; appeal flag set	terminal (re-application = new application)
WAITLISTED	Capacity-bound placement per priority factors	APPROVED (seat opens), REJECTED (cycle ends)

Status	Meaning	Next states
CONVERTED	Enrollment created; student exists (terminal)	-
WITHDRAWN	Parent withdrew during process	-
CANCELLED	Withdrawn/cancelled by school or expiry	-

Table 10-3: Document Status (per checklist item) and Approval Record

Domain	States / values	Rules
Document Status	PENDING > VERIFIED REJECTED > RESUBMIT	verified_by + verified_at mandatory on VERIFIED; OCR extract stored; resubmit loop unlimited within admission window
Approval Record (per level)	decision: APPROVED REJECTED WAITLISTED DEFERRED; levels: 1 Branch (Principal), 2 Academic (Coordinator), 3 Finance (Owner), 4 Board-tier (chain policy)	Levels configured per tenant; fee-waiver amount routes level (BRC R-APR-001); every decision stores reason + decided_at; parent-visible only as final decision
Enrollment Status	OFFERED > ACCEPTED > FEE_PENDING > ENROLLED	Offer expiry 7-15 days (PRD R-ADM-005; expiry -> CANCELLED); FEE_PAYMENT gates ENROLLED (ADR-037 FEE_PAYMENT -> ENROLLED)

CANONICAL RULE (RULE 7) Lead Status is a CRM state; Application Status is a processing state; Document Status is a checklist state; the Approval Record is a decision log; Enrollment Status is a contract state. A lead can be QUALIFIED while its application is DOCUMENT_PENDING; a funnel report computes stages from activities, never from statuses alone.

10.2 Canonical Pipeline

```

LEAD (NEW) -> CONTACTED -> QUALIFIED
  | activities: call, WhatsApp, visit scheduled, visit completed, counselling
  | session
  v
CONVERT: create APPLICATION (from lead or walk-in; lead_id NULL for walk-in)
  v
DRAFT -> SUBMITTED -> DOCUMENT_PENDING
  |   per-item: PENDING -> VERIFIED / REJECTED -> RESUBMIT ->
  v
DOCUMENT_VERIFIED -> UNDER_REVIEW

```

```

    | counselling session (SCHEDULED/COMPLETED/NO_SHOW) logged; recommendation
    recorded
    | approval chain per level: APPROVED / REJECTED / WAITLISTED / DEFERRED
    (reason mandatory)
    v
    APPROVED --> offer (fee quote valid 7-15 days)
    REJECTED -> rejection reason + parent feedback + appeal_eligible flag
    WAITLISTED -> priority queue (sibling, staff-child, alumni, distance...) [BRC
    R-ELG-011]
    v
    ENROLLMENT: OFFERED -> ACCEPTED -> FEE_PENDING -> ENROLLED
    | on ENROLLED: student created, admission number issued, guardian linked,
    | class assigned, fee plan assigned (see Sec. 11)
    v
    student ACTIVE (admissions complete; lifecycle continues in Sec. 12)

```

Workflow SLAs are adopted from ADR-037 where they add operational value without changing the status model: parent confirmation reminder at 7 days with escalation at 14; internal approval task reminder at 3 days with escalation to Principal; stalled-application metric flagged when UNDER_REVIEW exceeds 7 days. Age eligibility is validated at submission and re-checked at approval (AgeEligibilitySpecification; BRC age bands per program); capacity is checked at approval with waitlist offered when the class is full (AC-010).

10.3 Duplicate And Edge Handling

- Duplicate lead: phone (+school) unique index; on create, matching leads are surfaced and merge is offered; merged lead becomes DUPLICATE with pointer to survivor.
- Duplicate application: one open application per child per program per academic year; second submission returns 409 ADMISSION_DUPLICATE with link to existing.
- Public-form spam: rate-limited per IP + captcha; unverified applications auto-cancel after the admission window.
- Mid-year admission: admission_type MID_SESSION; prorated fee quote generated from fee rules.

11. Enrollment Flow

Enrollment converts an approved admission into an operating student record. It is the single most cross-cutting transaction in the product (admissions, student, finance, communication domains all participate), which is why the sources describe it both as a single transaction and as a saga. The canonical ruling: the enrollment record itself is created synchronously in one

transaction; the fee-plan assignment, first invoice generation and welcome communications run as follow-up steps through the outbox, with the workflow engine tracking completion.

```

    TRIGGER: application APPROVED + offer ACCEPTED (parent confirms; or office
    confirms on behalf)
    v
    [1] CREATE ENROLLMENT (one tx)
        validates: application APPROVED, no existing enrollment for (student-
        context, academic year),
                class capacity, age eligibility re-check
    v
    [2] STUDENT CREATED (same tx)
        status PROVISIONAL; admission number STU-YYYY-NNNNN issued (unique per
        tenant, immutable)
        student_history: ADMITTED event with full snapshot
    v
    [3] GUARDIANS LINKED (same tx)
        >= 1 guardian mandatory (from application parent info); is_primary,
        is_fee_payer,
        can_pickup, custody fields set; guardian dedup by phone+tenant
    v
    [4] ASSIGNMENTS (follow-up steps, workflow-tracked)
        class/section assigned -> classroom_assignment (unique student+academic
        year)
        fee plan assigned      -> fee plan per program; installment schedule
        generated
        transport assigned     -> optional (v1.1); route + stop mapped
        first invoice generated -> invoice DRAFT -> ISSUED; fee reminder schedule
        armed
    v
    [5] ACTIVATION GATE
        FEE_PENDING -> first installment paid (or approved waiver/installment
        plan)
                                -> student ACTIVE; parent portal credentials issued (Sec.
        13)
        non-payment in window -> enrollment stays FEE_PENDING; reminder cadence
        runs;
                                Principal may extend or cancel (-> CANCELLED,
        application -> WITHDRAWN)
    v
    events: EnrollmentCreated, StudentCreated, PaymentReceived, welcome message to
    guardians
    audit: every step recorded (actor=system for automated steps)
  
```

If any follow-up step fails (invoice generation, communication), enrollment is NOT rolled back; the workflow engine retries with backoff and raises an operational alert after the DLQ threshold, preserving the eventually-consistent UX copy rules of ADR-035 (the UI shows Enrollment confirmed - invoice pending rather than an error).

12. Student Lifecycle

The student is the longest-lived entity in PreOne and the one most tightly regulated (child data, DPDP retention). The canonical lifecycle below merges the four source variants, resolves the graduation-versus-promotion conflict, and defines the five exit operations (archive, deactivate, withdraw, transfer, delete) as semantically distinct actions as the brief demands.

Table 12-1: Canonical student states

State	Meaning	Entry	Exits
PROVISIONAL	Enrollment created; first installment pending; not yet attendable	Enrollment step 2	ACTIVE (payment/waiver), WITHDRAWN
ACTIVE	Attending; present in class rosters, attendance, finance	Payment confirmed	INACTIVE, GRADUATED, TRANSFERRED, WITHDRAWN, ARCHIVED
INACTIVE	Temporarily not attending (long illness, suspension of services, fee hold); roster-excluded	Admin action with reason	ACTIVE (reinstate), WITHDRAWN, ARCHIVED
GRADUATED	Completed the top program; alumni record; retention clock variant applies	Graduation event at year end	ARCHIVED
TRANSFERRED	Moved out: INTERNAL (other branch of tenant) or EXTERNAL (TC issued)	Transfer flow	ARCHIVED (after retention), or back to ACTIVE (internal reversal within window)
WITHDRAWN	Left mid-cycle by parent choice; dues settled per finance rules	Withdrawal request + clearance	ARCHIVED
ARCHIVED	Read-only, minimized dataset per DPDP; visible only in archive register	Retention/archival job or admin action with reason	Hard purge after retention (irreversible, job-only)

12.1 Promotion, Transfer, Withdrawal, Archive, Deactivate, Delete

Table 12-2: The five exit operations vs delete

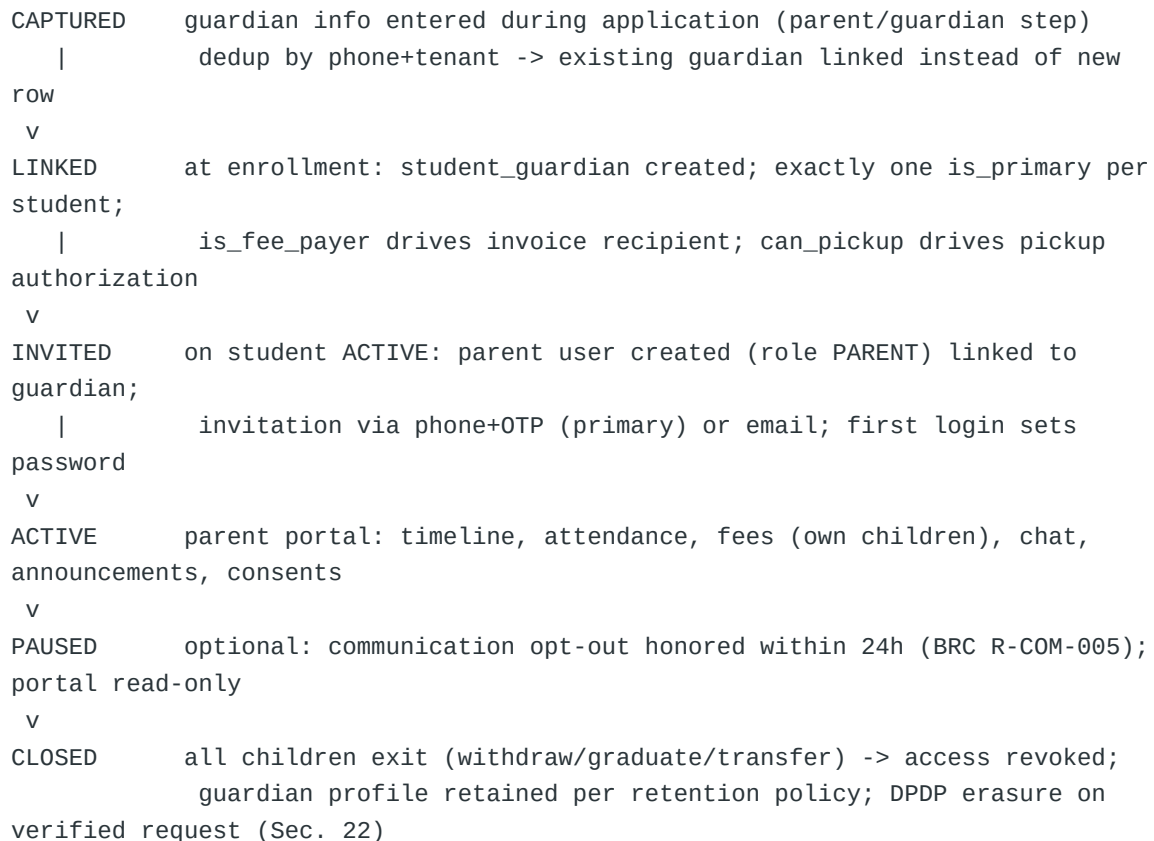
Operation	What it does	Reversible?	Who	Data effect
Promote	Event (not state): creates next-year classroom assignment + history entry at year boundary	Yes (assignment edit within window)	Principal	Student stays ACTIVE; history append-only
Transfer	INTERNAL: branch change within tenant (roster, transport, fees re-scope). EXTERNAL: TC generated, status TRANSFERRED	Internal: yes. External: no (re-admission is new)	Principal (internal); Owner (external)	History TRANSFERRED event; balances carried/squared per finance
Deactivate	Temporary INACTIVE with reason; excluded from rosters and reminders	Yes (reinstate keeps history)	Principal	No data removed; attendance stops
Withdraw	Permanent exit mid-cycle; dues cleared; parent access revoked	No	Principal + Owner countersign if dues outstanding	Financial records retained; profile retained
Archive	DPDP archival: minimize, read-only, retention_until set	No (purge only later)	System (retention job) / Admin with reason	PII minimized per archive_policy; archive register entry
Delete (hard)	FORBIDDEN in application code (ADR-047); reserved to retention job on ephemeral data only	-	System only	-

Every one of these operations is audited with before/after values and a mandatory reason, and every one emits the matching domain event (StudentTransferred, StudentArchived, ParentAccessRevoked, and so on) so that downstream modules (attendance rosters, fee reminders, transport lists, parent portal) update through the event pipeline instead of direct coupling. API surface: DELETE /students maps to archive-with-reason for admins (soft delete with 7-year DPDP retention), never to a SQL DELETE; the Show-deleted toggle and Restore affordance follow ADR-047.

13. Parent Lifecycle

No source document defines the parent lifecycle, although the data model is rich (guardian master + student_guardian link with is_primary, is_fee_payer, can_pickup, custody_percentage). This section defines the lifecycle normatively, using only mechanisms the sources already contain: the guardian model of the ERD, the parent-as-user identity model, and the communication consent framework.

13.1 Canonical Parent Flow



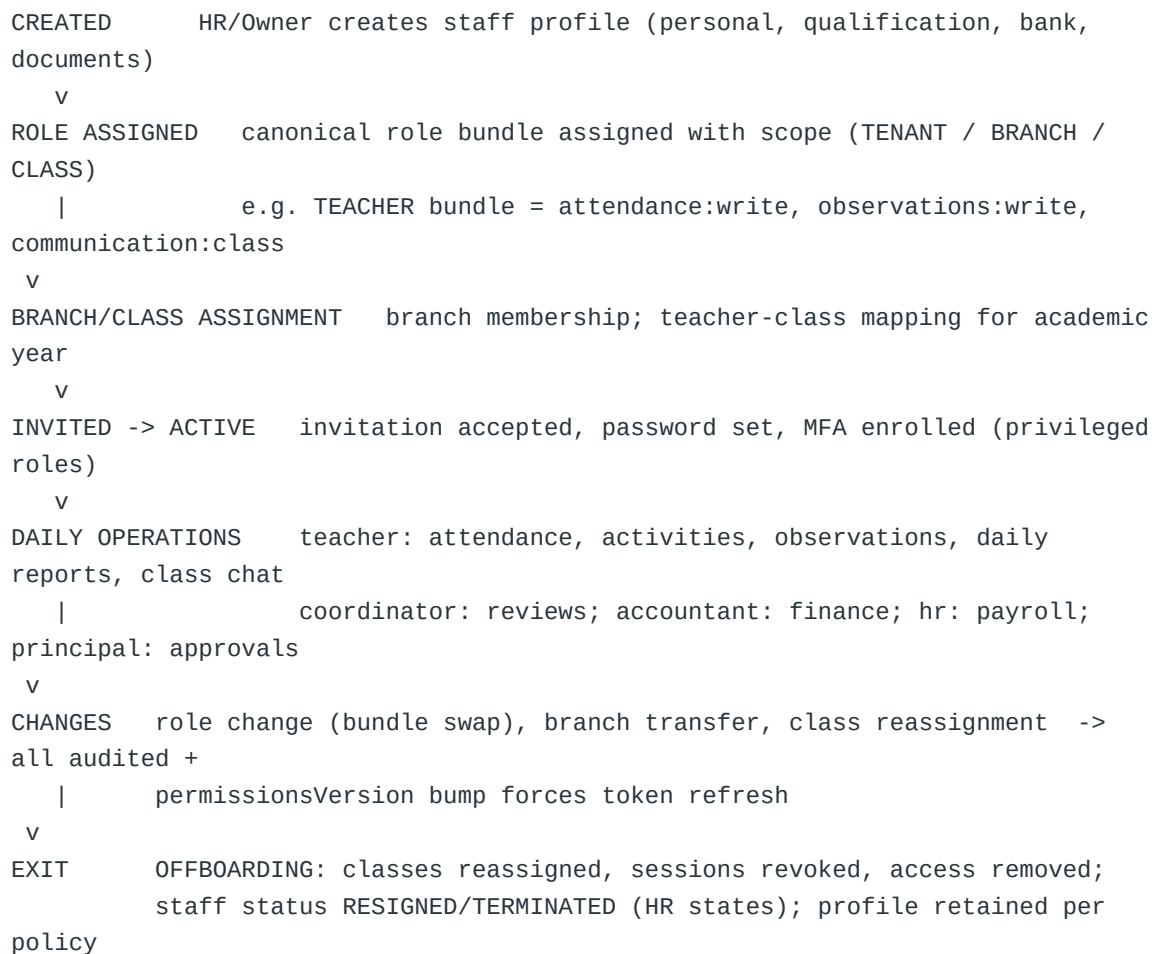
13.2 Guardian Rules

- Multiple guardians per student supported (up to reasonable limit); exactly one primary; one or more fee payers; up to 4 authorized pickup persons per student including non-guardians (BRC R-OPS-001), each with photo + ID proof.
- One guardian may be linked to multiple students, including across branches of the same tenant; the parent app shows a child switcher.
- Emergency contact is a guardian contact field flagged as emergency_primary; separate from pickup authorization.
- Communication preferences (language, channels, quiet hours) are per parent user; school messages respect them except emergency channel overrides (BRC R-COM-012).

- Custody disputes: custody_arrangement + court-order document flag restricts pickup and information visibility per flagged guardian; change requires Principal approval and is audited.

14. Teacher / Staff Lifecycle

Staff lifecycle in the sources is fragmented: the HR context owns staff records, the identity model owns users, and the only onboarding event is StaffOnboarded. The canonical lifecycle unifies them: every staff member is one user with one HR profile and one or more scoped role assignments.



14.1 What Teachers Can And Cannot Approve

- CAN: submit attendance; create/edit own activities and observations within the 24h window; draft daily reports; initiate parent chat; request attendance correction (not self-approve).
- CANNOT: approve admissions or any approval-level decision; approve attendance corrections (attendance:approve belongs to Principal/Coordinator tier); approve

refunds, discounts or fee waivers; publish report cards (Principal signs off); broadcast beyond their own class; delete anything.

- AI boundary: teachers may accept, edit or reject AI-drafted observations and daily sheets; their acceptance action is recorded with provenance (Sec. 20).

15. Attendance Flow

Attendance is the highest-frequency write in the product and the one parents watch in real time. The sources agree on purpose but conflict on statuses (3 vs 5), the edit model (edit-with-reason vs immutable-plus-correction) and escalation. The canonical flow adopts the five-status model, the append-only correction design, and the BRC escalation timers.

15.1 Canonical Marking Workflow

```
TEACHER opens Attendance (tablet) > select class > select date (today default;
future dates BLOCKED)
  v
SYSTEM loads roster (ACTIVE students of class for that day, incl. INACTIVE
exclusions)
  v
MARK: one-tap per student  PRESENT | ABSENT | LATE | HALF_DAY | LEAVE
      + optional check-in time, arrival mode, reason_if_absent, note
      bulk default PRESENT with per-student override (attendance_bulk snapshot
per class-day)
  v
SUBMIT (attendance:write)
      validation: one row per student per date (unique index); student ACTIVE
in class;
      late if arrival > start + 30 min auto-suggested (BRC R-OPS-004)
  v
PARENT NOTIFY: absence alert within 15 min (Push+SMS); no-ack in 30 min -> call
primary parent
                (BRC R-NOT-002; overrides PRD instant-only wording)
  v
LOCK-FREE CONTINUITY: same-day edit allowed with reason (original kept in audit
trail);
      next-day onward: CorrectionRequest (from_status, to_status, reason) ->
      approved by attendance:approve (Principal/Coordinator tier) -> applied +
audited
  v
REPORTS: monthly attendance_summary computed (present/absent/late/half/leave
days,
      attendance_percentage); promotion eligibility reads < 75% flag (BRC R-
OPS-005)
```

15.2 Statuses And Rules

Table 15-1: Attendance statuses (canonical, resolves C08)

Status	Counts as present?	Parent notify	Notes
PRESENT	Yes	No	Default bulk value
ABSENT	No	Yes (15 min)	reason_if_absent encouraged; leave may pre-explain
LATE	Yes (flagged)	Yes (info)	arrival_time stored; > 30 min past start auto-suggested
HALF_DAY	Partial	Yes (info)	check-out time distinguishes early-departure cases
LEAVE	No (excused)	No	StudentLeaveRequest types: casual, sick, medical, family, vacation, bereavement

- Escalation: teacher has not marked by start + 30 min -> system reminder; + 60 min -> Coordinator; + 120 min -> Branch Admin may mark on teacher's behalf (BRC R-OPS-006), attribution preserved.
- Historical integrity: past-date marking by a teacher requires the same CorrectionRequest path; system imports (biometric/RFID) are source-tagged (MANUAL, BIOMETRIC, RFID, APP).
- Fee-hold interaction: when a >60-day defaulter service-hold is active, attendance marking for that student stays available (attendance is a welfare record) but ID-card and optional services are blocked (BRC R-FIN-002 note).
- Audit: every create, same-day edit and correction writes an audit row; the monthly summary records computed_at and the correction count it absorbed.

16. Activities Flow

The learning-development chain is Activity > Observation > Milestone assessment, with Portfolio as the curated artifact layer. The sources define each concept differently or not at all; the canonical definitions below are extracted from the ERD tables (activity, observation, milestone, portfolio), the DDD invariants (edit windows, AI provenance) and the BRC quality rules, with the frequency conflict resolved conservatively.

Table 16-1: Concept definitions (never interchangeable)

Concept	Definition	Created by	Edited by	Parent sees
Activity	A curriculum unit or experience conducted with a class or child (ERD activity; lesson_plan link)	Teacher / Coordinator (library)	Owner of record within 24h	Timeline entry (auto within 5 min per BRC R-OPS-017)
Observation	A teacher's structured note about one child tied to activity context and learning outcomes; may be AI-drafted (ai_assisted flag)	Teacher	Within 24h (DDD); corrections = new linked observation after window	Only when shared (is_shared_with_parent, default true)
Milestone assessment	Per-student rating of a catalog milestone: NOT_STARTED, EMERGING, ACHIEVING, MASTERED, CONCERN	Teacher	Superseded by new assessment (history kept)	Milestone progress view + concern alerts (R-ACD-007)
Portfolio item	Curated artifact (photo/artwork/video) with category and optional AI caption	Teacher	Reorder/recaption anytime	Published portfolio only
Achievement	Discrete award (academic, sports, arts) with certificate	Teacher / Principal	-	Immediately on publish
Daily report	End-of-day summary (meals, nap, mood, toileting) merged from operational logs	System-assisted (teacher confirms)	Until daily-send	Daily report within 30 min of pickup (R-NOT-003)

16.1 Permissions And Quality

- Creation: teachers create activities, observations and assessments for their assigned classes only; coordinators create for any class in their scope; nobody edits another teacher's observation (comments instead).
- Quality loop: system flags low-quality text (<20 chars or generic words) per BRC R-ACD-005; Coordinator reviews weekly and recurring low quality triggers coaching; flag thresholds configurable.

- Frequency ruling (C14): system minimum 1 observation per child per week (alert to teacher, then Coordinator); dashboard target 2/week; PRD's 3/week is aspirational and not enforced.
- Attachments ruling (C15): 5 attachments max, 5 MB each (DDD wins); photos auto-tagged with consent check (media-use consent per child; non-consenting children excluded or AI face-blurred per R-OPS-016).
- AI drafts: allowed for observations, daily sheets, report-card narrative; always `ai_draft=true` until a teacher accepts/edits/rejects; rejection records feedback for model tuning (no PII in training).

17. Growth Passport Flow

AUDIT FINDING (UNDEFINED IN SOURCES) The term Growth Passport appears in downstream and implementation materials but is defined in ZERO of the 19 source documents: no entity, no flow, no permission, no report. The nearest source concepts are Portfolio (ERD), the parent Timeline (PRD communication domain), Milestone assessments and Report Cards. Per the audit rules this is declared UNDEFINED and the composition below is marked RECOMMENDED pending product sign-off (Sec. 31, OD-1).

17.1 Proposed Canonical Composition (RECOMMENDED)

Growth Passport is defined as the longitudinal, parent-facing record of a child's development, composed mechanically from existing records rather than being a new data silo: milestone assessments over time, published observations, portfolio items, achievements, activity participation and term report cards, plus AI-generated growth narratives and delay alerts where configured.

```

STUDENT (ACTIVE)
  | daily:    activities + daily logs          -> Timeline entries
  | weekly:   observations (shared)            -> Learning stories
  | periodic: milestone assessments           -> Skill progression
(EMERGING>MASTERED)
  | curated:  portfolio items                  -> Artifacts
  | events:   achievements, incidents summary -> Highlights
v
GROWTH PASSPORT VIEW (parent app + web)
  | filters:  academic year, domain
(cognitive/motor/social/language/emotional), class
v
GROWTH PASSPORT REPORT (term PDF)
  | compiled at term end: narrative (AI draft -> teacher review), milestone
map,
  | attendance %, portfolio highlights, teacher + principal sign-off
v
published to parent (report card approval chain applies, Sec. 16)

```

17.2 Permissions And Data Rules (RECOMMENDED)

- Read: parents see only their child's passport; teachers read/write for their classes; coordinators and principals read school-wide; export generates the term PDF.
- Write: no module writes directly into a passport; everything is projected from source records, so the passport can be rebuilt at any time (single source of truth preserved).
- AI: narrative drafts carry ai_draft provenance and require teacher acceptance before publish; AI never modifies milestone ratings or observations (Sec. 20).
- Retention: passport follows student record retention; on archive, parent access revoked and minimized export issued once.

18. Fees & Finance Flow

Finance has the richest rule set in the BRC and the most conflicts between PRD, DDD and the data docs. The canonical lifecycle is straightforward; the rulings below fix precision, GST, late fee, refunds and numbering. Financial transactions are always auditable (audit brief RULE 11), never hard-deleted, and every mutation flows through the idempotency-protected payment API.

18.1 Concepts And Lifecycle

Table 18-1: Finance concept definitions

Concept	Definition	Key fields / states
Fee Head	Catalog line type: tuition, admission, transport, meal, activity, exam, library, late-fee, other	type FIXED/VARIABLE/RECURRING/ONE_TIME; frequency; taxable flag
Fee Structure / Plan	Template of heads + amounts + frequency for a program (branch-optional scope), with security deposit and installment count	total_annual_cents; installments 1/2/4/12
Student Assignment	Plan applied to a student at enrollment; concession stack (sibling, staff-ward, scholarship, hardship) attached	fee_concession rows; approval trail
Invoice	Billing document generated per student per cycle (or on-demand)	DRAFT > ISSUED > PARTIALLY_PAID > PAID OVERDUE CANCELLED WRITTEN_OFF; immutable after ISSUE
Payment	Money received against invoice(s) or as advance	PENDING > SUCCESS FAILED; REFUNDED; gateway ref + Idempotency-Key

Concept	Definition	Key fields / states
Receipt	Proof document issued on successful payment; sequential per branch per FY	RCP-YYYY-NNNNN; PDF stored; reprint = copy flag
Refund / CreditNote	Money returned (against payment) or invoice correction without money movement	REQUESTED > APPROVED > PROCESSED REJECTED CANCELLED
Ledger	Double-entry journal projection (Prisma JournalEntry) for reporting and GST returns	Derived; never hand-edited

FEE PLAN (structure) -> assigned to STUDENT (concessions applied, approved per tier)

v

INVOICE GENERATED (schedule: monthly/term/annual per plan) DRAFT

v ISSUED (immutable; number INV-YYYY-NNNNN unique per school per FY; parent notified D-7)

v

PAYMENT (online gateway OR offline record; Idempotency-Key; cash < INR 50k rule, PAN capture)

| allocations across invoices (payment_allocation); part-pay allowed -> PARTIALLY_PAID

v

RECEIPT issued within 60 seconds of success (RCP-YYYY-NNNNN per branch per FY)

v

STATES: PAID | PARTIALLY_PAID | OVERDUE (due date passed) | CANCELLED | WRITTEN_OFF (2-tier approval)

v

LATE FEE: BRC slabs on overdue (grace 7d): 8-30d INR 100; 31-60d INR 500; >60d INR 1,000

+ service hold (ID card, optional services) with Principal approval

v

ESCALATION REMINDERS: D-7 Push; D-3 Push+SMS; overdue Push+SMS+Email+WhatsApp; auto-stop on ack

v

REFUND: REQUESTED (Accountant) -> APPROVED per tier -> PROCESSED ->

Receipt/Invoice adjusted;

REJECTED with reason; eligibility window 90 days (RECOMMENDED); audit every step

18.2 Rulings Applied To Conflicts

- Money precision (C10): all amounts are integer paise in storage and transit; UI formats INR with 2 decimals; exports carry paise with a documented display rule. The ERD 3.1 numeric(12,2) sentence is void.
- GST (C11): tuition/activity/exam for preschool = 0% (exempt) per BRC R-FIN-007; transport/meal/materials taxed per HSN as tenant-configured; invoice_tax

stores CGST/SGST/IGST breakup when applicable. PRD's flat 18% is void for tuition.

- Late fee (C12): BRC slabs, configured in late_fee_rule per tenant (grace default 7). DDD's per-day model dropped.
- Refunds (C13): initiation by ACCOUNTANT; approval tiers mapped to canonical roles - Principal <= INR 5,000; School Owner <= INR 25,000; Owner + finance committee above; TAT 14 working days; eligibility window 90 days from payment (RECOMMENDED merge); refunds > INR 5,000 require step-up MFA.
- Discount caps: stack total <= 25% default cap; above requires the approval tiers of BRC R-APR-007; sibling/staff/early-bird percentages per BRC R-FIN-005/006/R-ELG-012.
- Invoice mutability (C25): PATCH only in DRAFT; after ISSUE all corrections via CreditNote or Refund; middleware blocks UPDATE on issued invoices.
- Cash rule (BRC R-FIN-010): cash rejected > INR 50,000 with PAN capture below; digital mandatory above.

19. Communication Flow

Communication spans three mechanics: 1:1 chat (parent-teacher), announcements/broadcasts (school-to-many), and event-triggered notifications (system-to-person). The canonical model merges the ERD announcement/chat family with Prisma's Message/Conversation into one pipeline: Event > Notification Engine > Channel > Delivery > Read > Audit.

19.1 Who Can Message Whom

Table 19-1: Messaging permission matrix

From > To	Parents (own child)	Teachers	Staff	All parents (broadcast)
Parent	n/a	Chat: own child's teachers only	No	No
Teacher	Chat: own class parents; reply within SLA	Staff groups	Staff groups	No (class broadcast only, own class)
Coordinator	Via teacher; class-level broadcast own classes	Yes	Yes	Class-level only (BRC R-COM-004)
Principal	Chat + announcement (school)	Yes	Yes	Yes (self-approved per PRD; approval step skipped for sender)

From > To	Parents (own child)	Teachers	Staff	All parents (broadcast)
School Owner	Announcements	Yes	Yes	Yes
Accountant	System fee notices only (templated)	No	Yes	Fee-category broadcast only
System (events)	Transactional notifications	As configured	As configured	As configured

19.2 Channels And Delivery Pipeline

- Channels: In-App (always), Push, SMS, WhatsApp, Email; IVR call added for emergency escalation only (BRC R-COM-012). Provider routing per tenant with fallback chains (SMS fail -> email -> push per ADR-009).
- Pipeline: domain event -> outbox -> Redis Stream -> notification_rule evaluation -> template render (per language preference) -> channel queue (BullMQ) -> provider -> delivery webhook -> read tracking -> audit.
- Rate limits and quiet hours: WhatsApp <= 5 messages/parent/day; push 10/h; SMS 5/h; email 20/h; quiet hours 22:00-07:00 except CRITICAL priority.
- Chat rules: 1:1 staff-parent conversations; messages append-only with 5-minute edit window; conversations archive after 1 year; opt-out parents are not addressable except emergencies.

19.3 Event-Triggered Notification Catalog

Table 19-2: Canonical trigger catalog (merged from BRC R-NOT-001..012 + domain events)

Event	Audience	Channels	Timing / SLA
Fee due approaching	Fee payer	Push (D-7), Push+SMS (D-3), on-due reminder (RECOMMENDED)	Scheduled per plan
Invoice overdue	Fee payer	Push+SMS+Email+WhatsApp	On overdue + cadence; auto-stop on acknowledgement
Child marked absent	Primary guardian	Push+SMS	Within 15 min; no-ack 30 min -> IVR call
Daily report ready	Guardians	Push / In-App	Within 30 min of pickup
Observation shared	Guardians	Push / In-App	On publish
Milestone concern flagged	Guardians	Push + email	Same day as assessment
Incident MEDIUM	Primary guardian	Push+SMS	Within 30 min
Incident HIGH/CRITICAL	Primary guardian + Principal	Immediate call + Push+SMS	Immediate

Event	Audience	Channels	Timing / SLA
Admission approved / enrolled	Guardians	Email + Push	On event
Student transferred / exited	Guardians	Email	On event
Broadcast / announcement published	Audience per type	Configured channels	Approval first (R-COM-008)
Event reminder (school event)	RSVP audience	Push	14-day, 7-day, 1-day reminders
Holiday declared	All	Push+SMS	48 h prior
Transport delay > 10 min (v1.1)	Route parents	Push+SMS	Immediate
Salary credit (staff)	Employee	Email+Push	On payroll disbursement
Security (new device, password change, MFA change)	The user	Email + Push	Immediate

20. AI Center Flow

AI in PreOne is an intelligence layer that reads validated operational data and proposes assistance; it is never the system of record and never writes core records. The sources name six assistant capabilities (observation assistant, daily sheet auto-draft, report-card paragraph generation, fee-default prediction, parent reply composer, growth milestone prediction), the API catalog exposes five endpoints with quotas, and the DDD defines the AI Gateway with tenant quotas, audit and safety filters. What is missing everywhere is persistence and the human-review state machine; this section defines both.

20.1 Canonical AI Pipeline

OPERATIONAL DATA (attendance, observations, fees, admissions, milestones - tenant-scoped)

v

VALIDATION LAYER (only consented, non-archived, validated records; PII minimized per purpose)

v

AI GATEWAY (quota per tenant per feature; cost tracking; safety filters; provider ACL)

| providers: Z.ai GLM primary; OpenAI reserve; vision model for photo tasks (school-internal)

v

AI INSIGHT / DRAFT (artifact persisted in ai_insight with provenance)

v

HUMAN REVIEW (accept / edit / reject by authorized role; rejection stores feedback)

v

ACTION APPLIED (becomes normal operational record, author = the human who accepted)

v

AUDIT (ai_insight id, model, version, prompt hash, reviewer, decision - full trail)

20.2 Capability Catalog And Review Requirements

Table 20-1: AI capabilities (merged PRD + API catalog + BRC)

Capability	Input	Output	Human review	Quota (per tenant/day)
Observation assistant	Activity context + child + outcome tags	Draft observation text	MANDATORY teacher accept/edit/reject (ai_draft until then)	1,000 (mini model)
Daily sheet auto-draft	Day's operational logs (meals, nap, mood)	Draft daily report	MANDATORY teacher confirm	Included in observation quota
Report card narrative	Milestones + observations + attendance	Paragraph draft	MANDATORY teacher review + Principal sign-off chain	200
Parent reply composer	Incoming parent message + context	Draft reply	MANDATORY staff send decision	500
Lesson plan draft	Program + theme + outcomes	Plan draft	MANDATORY teacher edit	50
Fee default prediction	Payment history + attendance (aggregate)	Risk score + ranked list	Advisory only; no automatic action; Accountant/Principal view	20
Attendance risk / milestone delay alerts	Trend aggregates	Alert + confidence	Advisory; routes to Principal dashboard	20
Growth narrative (passport)	Passport projections	Narrative draft	MANDATORY teacher + Principal (report chain)	100

20.3 Classification Of Generated Content

- AI-generated: any artifact with ai_draft=true and no reviewer action yet; visually badged AI DRAFT everywhere it renders; excluded from reports and parent surfaces.
- Human-approved: AI artifact after accept/edit; provenance retained (model, version, reviewer, timestamp) forever; author of record is the human.

- System-generated: deterministic content (summaries, aggregations, computed percentages); labeled system, no review needed, still auditable.

20.4 Persistence And Guardrails

- New tables (RECOMMENDED, gap C31): ai_insight (id, tenant_id, feature, model, model_version, prompt_hash, input_refs, output_text, status DRAFT|ACCEPTED|EDITED|REJECTED, reviewer_id, reviewed_at, related_entity_type, related_entity_id) and ai_usage (tenant_id, feature, tokens_in, tokens_out, cost_paise, latency_ms, created_at) for the AI Gateway dashboard (tokens, calls, latency, cost).
- Never-write list: AI cannot modify student master data, attendance rows, invoices, payments, approvals, roles or permissions; it can only create drafts or advisory insights (audit RULE 10).
- Consent: child media AI tasks require media-use consent; non-consenting children are excluded or face-blurred (BRC R-OPS-016); training on tenant data is opt-in and k-anonymized ($k \geq 10$).
- Anomaly detection on auth/permission/PII access remains a SECURITY capability (Security Sec. 14) - deterministic rules first, ML later; same audit trail.

21. Reporting Flow

Reporting consumes validated operational data through a read path that never touches write endpoints. The canonical pipeline and the report catalog merge the PRD's five report categories with the UI module list and the reporting database strategy of ADR-060 (replica for real-time, materialized views for dashboards, OLAP for cross-tenant analytics).

```
DATA (operational stores + read replicas)
  v REPORT QUERY (parameterized, whitelisted sorts/filters)
FILTER (scope: tenant > branch > class > year; role-based data scope R-RPT-001)
  v AGGREGATION (materialized views hourly; real-time via replica < 5 s)
REPORT (category, period, scope)
  v EXPORT / VIEW / SHARE (PDF/Excel/CSV via Export Engine; scheduled email;
signed links 24 h)
    every export: audit LOGGED (actor, rows, filters) per DPDP data-export
rule
```

Table 21-1: Canonical report catalog

Category	Reports	Primary audience
Operational	Daily attendance summary; late-arrival log; incident log; visitor log; staff attendance	Principal, Coordinator

Category	Reports	Primary audience
Academic	Milestone progress per child; observation coverage; AI acceptance rate; report card (term)	Teachers, Parents (own child), Principal
Financial	Daily fee collection; defaulters list; refund summary; discount register; GST filing report; P&L summary	Accountant, Owner
Admissions (management)	Lead funnel conversion; admission TAT; branch performance comparison; enrollment trend; capacity utilization	Owner, Principal
Compliance	DPDP consent register; DSAR log; audit trail export; data retention status	Owner, PLATFORM_ADMIN, Principal (read)
Growth Passport	Term passport PDF; milestone domain trends	Parents, Teachers

- Permissions: every report declares required scope; cross-tenant reports exist only on the platform console for PLATFORM_ADMIN; sensitive exports (PII, finance) need step-up MFA and are watermarked.
- Custom report builder (FR-049): drag-and-drop dimensions/measures over whitelisted datasets; saved reports are role-scoped; scheduled runs use the report queue with sandboxed execution.
- Predictive analytics (FR-050): admission conversion and fee default predictions run in the AI pipeline (Sec. 20), surfaced inside these reports with confidence display.

22. Audit Flow

Audit is the strongest-consensus area of the sources and this section mostly formalizes rather than reconciles: every important mutation is auditable, the record is immutable, written in the same transaction as the business change, hash-chained, retained 7 years and queryable through a read-only console. The unified record below merges the three competing audit shapes (C30) into one superset.

Table 22-1: Canonical audit record

Field	Type	Notes
id	uuid v7	Immutable PK
occurred_at	timestampz	Server time; BRIN-indexed

Field	Type	Notes
actor_id	uuid (nullable)	Null = system action; actors registry includes users, services, system
actor_role_snapshot	jsonb	Roles + scope at action time (Security Sec. 11)
tenant_id / branch_id	uuid	Scope of the action
action	enum	CREATE, UPDATE, DELETE(soft), LOGIN, LOGOUT, LOGIN_FAILED, EXPORT, IMPORT, ACCESS, APPROVE, REJECT, PERMISSION_CHANGE, ERROR
entity_type / entity_id	string	Polymorphic target
old_values / new_values	jsonb	Diff per UPDATE; before-image for DELETE; after-image for CREATE
reason	text (nullable)	Mandatory for corrections, archival, overrides, admin deletes (RECOMMENDED formalization)
ip_address / user_agent / device_id	inet / text / text	From validated proxy chain
request_id / trace_id	string	Correlates with logs and traces
result	enum	SUCCESS FAILURE DENIED (denials audited with reason per PRD 7.3)
hash_prev / hash_self	char(64)	SHA-256 chain per tenant; nightly verification + alert on break

Table 22-2: Audited event catalogue (minimum set)

Domain	Events
Identity & access	Login success/failure, logout, lockout, password change/reset, MFA enroll/revoke, session revoke, role assigned/removed, permission bundle change, invitation sent/accepted
Tenant lifecycle	Tenant created/activated/suspended/resumed/archived, plan changed, branch created/closed
Admissions	Lead created/merged/converted/lost, application submitted, document verified/rejected/resubmitted, approval decision each level, offer sent/expired, rejection+appeal

Domain	Events
Student	Created, updated, class assigned, promoted, transferred (internal/external), deactivated/reinstate, withdrawn, archived, parent access granted/revoked
Attendance	Marked (bulk or single), same-day edit, correction requested/approved/rejected, escalation override
Finance	Fee plan changed, invoice issued/cancelled/written-off, payment recorded/failed, receipt issued/reprinted, refund requested/approved/processed/rejected, discount/waiver approved
Learning	Observation created/edited/shared, milestone assessed, portfolio published, report card generated/published, passport exported
Communication	Broadcast created/approved/sent, message sent (metadata), opt-out honored, template changed
AI	Insight generated, reviewed (accepted/edited/rejected), quota event
System	Config/feature-flag change, integration credential change, data export (any), DSAR request/fulfilment, retention purge

- Immutability: INSERT-only permissions in the audit schema; no UPDATE/DELETE grants to the application; WORM object-lock cold storage after 90 days; 7-year retention (financial 8 years per IT Act where applicable).
- PII discipline: audit rows mask PII fields unless actor holds unmask permission; every PII read (reveal action) is itself audited (Security PII Access Log).
- Query surface: /platform audit console for PLATFORM_ADMIN (all tenants, read-only) and school audit page per tenant; export = audited operation; SIEM stream via outbox for platform monitoring.

23. Navigation Architecture

The UI Design Philosophy defines the shell (64px sticky header with global search, notifications and profile; 240px sidebar grouped by section; mobile bottom nav of 5 items + More) but never enumerates the final menu tree, and the module lists differ across documents (13 vs 8 vs 12 vs 14). The canonical information architecture below uses the 14-module backend decomposition, grouped for humans, with role-visibility and page-type annotations. Items marked [P] are platform-console only; [S] setup-only; [R] reporting; the rest are operational.

23.1 Final Information Architecture

PLATFORM CONSOLE (PLATFORM_ADMIN only)

Tenants [P] > registry, create wizard, activation checklist, suspend/offboard
 Subscriptions & Billing [P] | Feature Flags & Plans [P] | Platform Audit [P]
 | Integrations [P]

ADMIN PORTAL (school roles; items filtered by role + scope)

- Dashboard (KPI tiles per role; setup-progress card until complete)
- Admissions > Leads (list, capture) | Follow-ups & Visits | Pipeline board
- Students (view) > Applications | Documents queue | Waiting list > Student list | Profiles | Attendance (view) | Fees
- People (v1.1 custom) > Growth Passport | Documents > Parents & Guardians | Teachers & Staff | Roles
- Academics > Programs & Classes | Timetables | Lesson plans | Report cards
- Operations > Attendance (mark/correct) | Daily log | Incidents | Visitors & gate
- Finance > Fee structures | Invoices | Payments & receipts | Refunds
- > Discounts & scholarships | Defaulters | Finance reports [R]
- Communication > Announcements & broadcasts | Chat | Templates | Delivery logs
- Growth Passport > Child passports | Term compilation [R]
- AI Center > Drafts review queue | Insights | Usage & cost [R]
- Reports [R] > Operational | Academic | Financial | Admissions | Compliance
- Setup [S] > Academic years | Branches | Classes | Fee setup | Staff & roles
- > Approval workflow | Integrations
- Settings > Branding | Notifications | Security (MFA, sessions) | Consents
- > Billing (PreOne) | Onboarding progress

PARENT APP (PARENT only)

- Home (child switcher) | Timeline & Daily report | Growth Passport | Attendance
- Fees & Payments | Announcements & Events | Chat | Consents & Profile

Role-dependent visibility follows the security rule that navigation adapts to effective permissions: a user without reports:view never sees Reports; action buttons hidden without their permission; critical items (Dashboard, Settings, Logout) can never be hidden by navigation customization. Teacher tablets default to a task-focused home (Mark attendance, Today's activities, Daily report) implemented as a dashboard layout of the same app rather than a separate build.

23.2 Page-Level Specifications (Core Pages)

Using the source Student Profile specification as the quality benchmark, the table below fixes route, roles, key actions, validations, primary APIs and entities for the twelve core pages. Every page follows the shared anatomy: PageHeader (title + actions), filter bar, content cards (max 6 cards + 1 table + 1 chart per screen), pagination, and standard loading (skeleton), empty (illustration + CTA), error (toast + retry + traceId) states.

Table 23-1: Core page specifications

Page (route)	Roles	Entry points	Key actions	Validations	Primary APIs / entities
Login (/login)	All	Direct, guarded redirects	Password login; OTP login (parent); forgot password	Rate limit 5/min; lockout at 5 fails; generic errors	POST /auth/login, /auth/refresh; user, session
Platform Tenants (/platform/tenants)	PLATFORM_ADMIN	Console nav	Create tenant (5-step wizard), activate, suspend, offboard, impersonate-with-reason	Unique school name/email; plan valid; branch code unique	/platform/tenants; tenant, branch, user
Owner Onboarding wizard (/app/onboarding)	SCHOOL_OWNER	First login redirect	9-step checklist (Sec. 07); resume anytime	Step gates per Table 7-1	settings + academics + identity APIs; academic year, classroom, staff
Dashboard (/app/dashboard)	All school roles	Default landing	KPI tiles (per role), 7-day trends, quick actions, alerts feed	Scope filters honor branch switcher	/dashboard aggregate; many (read-only)
Leads (/app/admissions/leads)	COUNSELLOR, PRINCIPAL	Admissions nav	Capture lead, call/WhatsApp log, schedule visit, qualify, convert, merge duplicates, lost-with-reason	Phone unique per school; lost reason mandatory	/crm/leads(+follow-ups, convert); lead, lead_activity
Pipeline (/app/admissions/pipeline)	COUNSELLOR, PRINCIPAL	Admissions nav	Kanban by stage (funnel projection); drag = activity log, not status change	Stage moves validated against lead status	/crm/leads?view=funnel; lead, lead_activity
Applications (/app/admissions/applications)	COUNSELLOR, PRINCIPAL, COORDINATOR	Admissions nav	Create (stepper), submit, upload docs, verify/reject docs, counsel log, approve/reject/wait list per level, offer	Checklist complete before UNDER_REVIEW; approval reason mandatory; capacity + age checks	/admissions/applications(+documents, approve, reject); application, application_document, approval
Student List (/app/students)	PRINCIPAL, COORDINATOR, TEACHER (class scope), ACCOUNTANT (fee fields)	Students nav, global search	Filter/search/sort, bulk export (audited), archive with reason (admin)	Export needs student:export; archive reason mandatory	/students; student, classroom_assignment

Page (route)	Roles	Entry points	Key actions	Validations	Primary APIs / entities
Student Profile (/app/students/{id})	Same as list + PARENT (own child, app)	List, search, timeline links	Tabs: Overview, Attendance, Growth (passport), Fees, Documents; actions per role (edit, assign class, transfer, deactivate, archive)	Field-level permissions (medical, Aadhaar masked); optimistic version check	/students/{id}+ sub-resources; student, student_guardian, attendance, observation, invoice
Attendance (/app/operations/attendance)	TEACHER (mark own class), PRINCIPAL/COORDINATOR (view/approve)	Operations nav, tablet home	Mark day (bulk+override), submit, same-day edit, correction queue approve/reject	One row/student/date; future blocked; correction reason mandatory	/attendance/mark(+bulk, corrections); attendance, attendance_correction
Finance - Invoices (/app/finance/invoices)	ACCOUNTANT, SCHOOL_OWNER	Finance nav	Generate cycle invoices, issue, record payment (idempotent), receipt, cancel, credit note	Draft-only PATCH; cash < 50k rule; allocations sum <= invoice balance	/finance/invoices, /payments, /receipts; invoice, payment, receipt
Communication - Broadcast (/app/communication)	PRINCIPAL (school), COORDINATOR (class), OWNER	Communication nav	Compose (audience, channels, schedule), approval step, send, delivery/read stats	Audience-permission matrix Table 19-1; quiet hours; WhatsApp rate cap	/communication/announcements, /broadcasts; announcement, broadcast_log
Settings - Security (/app/settings/security)	All staff (self), OWNER (tenant policy)	Settings nav	MFA enroll, sessions list + revoke, trusted devices, password change	Step-up MFA for sensitive ops; session cap 5	/auth/mfa, /sessions; session, refresh_token, audit_log

24. RBAC Matrix

This section defines the five distinct concepts the brief asks to separate, fixes the permission grammar, and provides the role-by-module action matrix that both the UI (menu/action visibility) and the backend (guard enforcement) consume from the same source. RBAC is primary with ABAC fallback for scope conditions (Security Sec. 8); UI filtering is an optimization, backend authorization is the truth (Frontend Sec. 10.3).

24.1 Concept Separation

Table 24-1: Concept definitions

Concept	Definition	Example
ROLE	Named bundle of permissions assigned to users, optionally scoped; system roles immutable, custom roles tenant-defined (v1.1) with no-escalation guard	TEACHER, PRINCIPAL

Concept	Definition	Example
PERMISSION	Atomic capability in grammar module:action; grouped into bundles; dangerous ones flagged	students:create, attendance:approve
USER	An authenticated identity (staff, owner, parent); may hold multiple roles with different scopes	Rekha (TEACHER @Branch A, CLASS Nursery-A)
SCHOOL TENANT	The isolation boundary owning all data, configuration and users for one school business	Sunshine Preschool (tenant)
BRANCH	Physical campus inside a tenant; scoping level for assignments and data partitioning	Sunshine - Koramangala (branch)

24.2 Permission Grammar (resolves C21)

- Canonical grammar: module:action, lowercase, colon separator; action set: create, read, update, delete(soft), export, approve, mark, send, manage, execute.
- Module set is the 14 canonical modules (Sec. 27.1). Examples: students:create; attendance:mark; attendance:approve; finance:read; finance:execute; admissions:approve; platform:manage.
- Scoping suffixes are NOT part of the permission string; scope comes from the assignment (TENANT/BRANCH/CLASS) and the request context. Field-level visibility (medical, Aadhaar, salary) is a separate field-policy layer, not a permission explosion.
- Migration: FRONTEND STUDENT_READ = students:read; SECURITY STUDENT.READ and ENG students.read map identically; decorators unified as @RequirePermissions(...).

24.3 Role x Module Action Matrix

Legend: C create, R read, U update, D delete(soft/archive), A approve, F execute/financial, X full management. Blank = no access. Scope column notes where access is narrower than tenant.

Table 24-2: Role x module matrix (canonical)

Module	PLATF ORM_A DMIN	SCHOO L_OW NER	PRINCI PAL	COORD INATOR	COUNSELOR	TEACHER	ACCOUNTANT	RECEPTIONIST	HR_MANAGER	TRANSPORT_MGR	PARENT
platform	X	-	-	-	-	-	-	-	-	-	-
identity	X (console)	X	R,U	-	-	-	-	-	C,R,U	-	R (self)
crm	R (console)	R	R,A(-)	R	X	-	-	C,R	-	-	-

Module	PLATF ORM_A DMIN	SCHOO L_OW NER	PRINCI PAL	COORD INATOR	COUNS ELLOR	TEACH ER	ACCOU NTANT	RECEP TIONIS T	HR_MA NAGER	TRANS PORT_ MGR	PAREN T
admissions	R (console)	R	A	R	C,R,U	-	-	C,R	-	-	R (own)
student	R (console)	X	X	R,U (scope)	C,R,U (pre-enroll)	R (class)	R (fee fields)	R (contact)	-	-	R (own)
academics	-	R	X	X	-	R,U (own class)	-	-	-	-	R (own)
attendance	R (console)	R	R,A	R,A (scope)	-	C,R,U (own, same-day)	-	-	-	R (v1.1)	R (own)
communication	-	X	X	C,R,U (class)	R,U (leads)	C,R,U (class)	R (fee notices)	-	R,U (staff)	R,U (transport)	R (own), U prefs
finance	R (console)	X	R + refund initiate	-	-	-	X	R + receipt reprint	R (payroll)	R (transport fees)	R (own), C (pay)
inventory	-	R	R	-	-	R (class materials)	R	-	-	-	-
hr	-	X	R	-	-	R (self)	-	-	X	-	-
administration	R (console)	X	C,R,U	R	-	-	-	C (visitors)	R	R (vehicles)	-
reports	X (cross-tenant)	X	X (school)	R (academic)	R (funnel)	R (own class)	X (finance)	R (visitors)	X (HR)	R (transport)	R (own child)
settings	X (platform)	X	R,U (academic)	-	-	U (self)	U (self)	U (self)	U (self)	U (self)	U (own prefs)

24.4 Approval Authority Map

Table 24-3: Approval workflows and tier owners

Approval	Tier 1	Tier 2	Tier 3+	Source
Admission decision	PRINCIPAL	COORDINATOR (academic level)	OWNER (finance-waiver level)	ERD approval levels 1-4 mapped to canonical roles
Refund	PRINCIPAL <= 5k	SCHOOL_OWNER <= 25k	OWNER + finance committee > 25k	BRC R-APR-002 remapped (C13)
Discount stack	ACCOUNTANT <= 5%	PRINCIPAL 5-15%	SCHOOL_OWNER 15-25%; committee > 25%	BRC R-APR-001
Fee waiver	PRINCIPAL <= 25%	SCHOOL_OWNER 25-75%	committee > 75% or > 50k	BRC R-APR-007
Attendance correction (past date)	COORDINATOR	PRINCIPAL (>= 7 days old)	-	Canonical (Sec. 15)
Announcement to all parents	PRINCIPAL (self-approve if sender)	-	OWNER for policy-sensitive	BRC R-COM-008 remapped

Approval	Tier 1	Tier 2	Tier 3+	Source
Write-off	ACCOUNTANT initiate	SCHOOL_OWNER	Owner + committee > 10k	DDD invariant + BRC
Leave (staff)	REPORTING MANAGER <= 2 days	HR_MANAGER > 2 days	PRINCIPAL (disputed)	BRC R-APR-003

25. State Machines

This section gives the explicit machine for every major entity: states, allowed actions per state, resulting states, and the transitions that are irreversible. Every transition is permission-gated per Sec. 24 and audit-logged per Sec. 22; invalid transitions return 409 with the current state in the error details.

25.1 School (Tenant)

```
DRAFT --commit--> PENDING_ACTIVATION --activate(gate pass)--> ONBOARDING --
checklist done--> ACTIVE
PENDING_ACTIVATION --trial path--> TRIALING --convert--> ONBOARDING ; TRIALING
--expire--> ARCHIVED
ACTIVE --suspend(payment/violation)--> SUSPENDED --resolve--> ACTIVE
ACTIVE|SUSPENDED --offboard(30d window)--> ARCHIVED --purge(retention done)-->
[IRREVERSIBLE]
```

25.2 User (Staff / Owner / Parent)

```
PENDING_VERIFICATION --accept invite + set password--> ACTIVE
ACTIVE --5 fails / admin--> LOCKED --cooldown/admin unlock--> ACTIVE
ACTIVE --deactivate--> INACTIVE --reactivate--> ACTIVE
ACTIVE|INACTIVE --delete(soft, reason)--> DELETED [no login; recoverable within
window; then purge]
any --password reset / role change--> sessions revoked (state unchanged)
```

25.3 Admission Lead

```
NEW --first outreach--> CONTACTED --fit confirmed--> QUALIFIED --application
created--> CONVERTED [terminal]
NEW|CONTACTED|QUALIFIED --drip--> NURTURE --re-engage--> QUALIFIED ; --
close(reason)--> LOST [terminal]
any --duplicate detected--> DUPLICATE [terminal, merged pointer] ; LOST
reason mandatory, audited
```

25.4 Application

```
DRAFT --submit(lock form)--> SUBMITTED --checklist open--> DOCUMENT_PENDING
DOCUMENT_PENDING --all docs VERIFIED--> DOCUMENT_VERIFIED --enter review-->
UNDER_REVIEW
UNDER_REVIEW --approve chain complete--> APPROVED --offer accepted--> CONVERTED
[terminal -> Sec. 11]
```

```

UNDER_REVIEW --reject(reason)--> REJECTED [terminal; appeal flag]
UNDER_REVIEW --capacity full--> WAITLISTED --seat opens--> APPROVED ; --cycle
end--> REJECTED
DRAFT --withdraw/cancel--> CANCELLED [terminal] ; APPROVED --offer expired (7-
15d)--> CANCELLED

```

25.5 Document (per checklist item)

```

PENDING --verify--> VERIFIED [terminal for cycle]
PENDING --reject(reason)--> REJECTED --parent resubmits--> RESUBMIT --re-
verify--> VERIFIED|REJECTED

```

25.6 Enrollment

```

OFFERED --parent accepts (<= expiry)--> ACCEPTED --invoice raised-->
FEE_PENDING
FEE_PENDING --first installment success / approved waiver--> ENROLLED [terminal
-> student ACTIVE]
OFFERED --expiry--> CANCELLED [terminal] ; FEE_PENDING --window exceeded +
Principal cancel--> CANCELLED

```

25.7 Student

```

PROVISIONAL --fee/waiver confirmed--> ACTIVE
ACTIVE --temporary exit(reason)--> INACTIVE --reinstate--> ACTIVE
ACTIVE --year end / completion--> GRADUATED [exit]
ACTIVE --transfer internal--> ACTIVE (branch rescope) ; --transfer external
(TC)--> TRANSFERRED [exit]
ACTIVE|INACTIVE --withdraw (dues cleared)--> WITHDRAWN [exit]
GRADUATED|TRANSFERRED|WITHDRAWN --retention/archive job or admin(reason)-->
ARCHIVED --purge--> [IRREVERSIBLE]

```

25.8 Parent

```

CAPTURED --student enrolled--> LINKED --invite accepted--> ACTIVE
ACTIVE --opt-out (non-emergency)--> PAUSED --opt-in--> ACTIVE
ACTIVE --all children exited--> CLOSED [access revoked; profile retained per
policy]
any --verified DPDP erasure--> ERASED [token destruction; IRREVERSIBLE;
certificate issued]

```

25.9 Attendance (day-state per student)

```

UNMARKED --mark/submit--> MARKED --same-day edit (reason)--> MARKED (edited,
audit kept)
MARKED --next-day+--> CORRECTION_REQUESTED --approve--> MARKED (corrected) ; --
reject--> MARKED
monthly summary COMPUTED (absorbs correction count; never blocks later
corrections)

```

25.10 Invoice / Payment

```
Invoice: DRAFT --issue--> ISSUED --part pay--> PARTIALLY_PAID --full pay-->
PAID
ISSUED|PARTIALLY_PAID --due passed--> OVERDUE --full pay--> PAID
DRAFT --cancel--> CANCELLED ; ISSUED --credit note--> CREDITED ; bad debt (2-
tier)--> WRITTEN_OFF
[IRREVERSIBLE: ISSUE (draft-only edits end), PAID (no mutation; refunds only),
WRITTEN_OFF]
Payment: PENDING --gateway--> SUCCESS | FAILED ; SUCCESS --refund full-->
REFUNDED ; partial -> PARTIALLY_REFUNDED
```

25.11 Growth Passport (RECOMMENDED)

```
EMERGING(record-level) -> term COMPILE -> REVIEW (teacher+principal) ->
PUBLISHED -> parent visible
PUBLISHED --amendment--> REVISION (v+1, audit kept) ; no deletion after
PUBLISHED
```

25.12 Notification

```
QUEUED --engine--> SENT --webhook--> DELIVERED --open--> READ
QUEUED|SENT --provider fail x3--> FAILED --retry policy--> DLQ [manual ops]
CANCELLED only before SENT (broadcast abort)
```

26. API Flow

Every feature follows one request path and one response envelope. The canonical stack merges the Backend TD pipeline with the API catalog conventions and fixes the envelope, code and pagination conflicts (C19, C20, C34).

26.1 Canonical Request Path

```
UI (Next.js) -> HTTPS -> API Gateway (TLS 1.3, WAF, rate limit)
-> Middleware: traceId, tenant extraction (JWT tid), request logging
-> JwtAuthGuard (401 on missing/invalid/expired)
-> PermissionsGuard (module:action + scope; 403 before any data access; 60s
perm cache)
-> ValidationPipe (DTO class-validator + Zod; 422)
-> Idempotency check (writes that need it; 409 on key reuse with different
payload)
-> Controller -> Application Service (tx boundary, audit hook) -> Domain
(aggregates, invariants)
-> Repository (Prisma, tenant middleware + RLS) -> PostgreSQL
-> Outbox insert (same tx) -> domain events -> Redis Streams -> workers
-> Response envelope { success, data | error, meta, traceId }
```

26.2 Response Envelope (resolves C19)

Success: { success: true, data: <resource | array>, meta: { page, pageSize, total, totalPages, hasMore, nextCursor? }, traceId }. Error: { success: false, error: { code, message, field?, details[] }, traceId }. The OpenAPI document is to be regenerated from the API catalog with this envelope; unwrapped resource responses are void.

26.3 Error Code Families (resolves C20)

Table 26-1: HTTP and code mapping

HTTP	When	Code family / examples
400	Malformed request body / JSON parse	REQUEST_MALFORMED
401	Missing, invalid, expired token; disabled account	AUTH_001, TOKEN_EXPIRED, TOKEN_INVALID, SESSION_REVOKED, ACCOUNT_DISABLED
403	Permission or scope failure; suspended tenant	PERMISSION_DENIED, SCOPE_VIOLATION, TENANT_SUSPENDED, TENANT_PENDING
404	Not found or cross-tenant existence hiding	STUDENT_NOT_FOUND, LEAD_NOT_FOUND, NOT_FOUND
409	Conflict: duplicates, state-machine violation, idempotency, version	DUPLICATE_ADMISSION_NUMBER, ATTENDANCE_ALREADY_MARKED, INVALID_STATE_TRANSITION, VERSION_MISMATCH, IDEMPOTENCY_CONFLICT
422	DTO / semantic validation failure	VALIDATION_001, DOB_FUTURE_DATE, FIELD_TOO_LONG, AGE_INELIGIBLE
423	Account locked	ACCOUNT_LOCKED (retry-after)
429	Rate limit exceeded	RATE_LIMIT_001 (Retry-After header)
503	Maintenance mode	MAINTENANCE_001

26.4 Conventions

- Paths: /api/v1/{module}/{resource} kebab-case plural (e.g. /api/v1/admissions/applications, /api/v1/finance/invoices); module prefixes resolve

C-path ambiguity; PUT forbidden; PATCH for partial updates; DELETE is soft-delete/archive.

- Pagination: page-based default (page, pageSize <= 100); cursor pagination mandatory for feeds > 10k rows (audit search, attendance history); filters eq/ne/gte/lte/between/in/nin/like; sort whitelist only; q = full-text on indexed fields.
- Idempotency-Key mandatory on: payments, refunds, invoice issue, admission submit, tenant create, file upload complete; 24h replay window; same key + same payload = cached response; different payload = 409.
- Rate limits: anonymous 60/min/IP; authenticated 600/min/user; service keys 6,000/min; auth login 5/min; sensitive ops 10/min.
- Versioning: URL-versioned; deprecation via Sunset header + 12-month support; breaking DTO changes get V2 suffix; workflow definition changes follow ADR-040 minor/major rules.
- Realtime: WebSocket channels /ws/chat, /ws/attendance, /ws/notifications, /ws/dashboard, /ws/location (v1.1); rooms keyed tenant:branch; reconnect backoff 1s..30s.
- Webhooks (merged set, C34): admission.approved, student.admitted, fee.paid, invoice.generated, attendance.marked, student.promoted, reportcard.published; HMAC-SHA256 signature header; at-least-once with retry backoff.
- File uploads: pre-signed URL flow (15-min URL), magic-byte + MIME whitelist, ClamAV scan, quarantine on failure, keys tenants/{tenantId}/...

27. Database Relationship Flow

The canonical schema merges the two documentation generations (ERD v3.0 school_id world and Prisma v3.0 tenantId world) into one model. The ruling: the ERD is the domain-complete baseline (it contains admissions depth, observation/milestone/portfolio, communication logs, finance detail that Prisma lacks), Prisma contributes the implementation conventions (tenant/branch discriminators, 7-column audit block, soft delete middleware, version columns). The merged model adopts ERD table coverage with Prisma conventions, snake_case plural tables, and the audit block on every business table.

27.1 Canonical Entity Catalog (Core)

Table 27-1: Core entities

Entity	PK / keys	Core relationships	Status column	Scope	Audit cols
tenant	id uuid7; email unique	1-N branch, user via tenant_user; 1-1 subscription	PENDING_ACTIVATION TRIALING ONBOARDING ACTIVE SUSPENDED ARCHIVED	-	yes
branch	id; (tenant_id, code) unique	N-1 tenant; 1-N classroom, staff assignment	ACTIVE INACTIVE SUSPENDED CLOSED	tenant	yes
academic_year / session	id; (tenant_id) one ACTIVE	1-N terms; partitions attendance, invoice, promotion	UPCOMING ACTIVE ARCHIVED	tenant	yes
user	id; email/phone identity	N-M role via user_role (scope TENANT/BRANCH/CLASS); 1-1 staff for employees	PENDING_VERIFICATION ACTIVE INACTIVE LOCKED DELETED	platform-global identity + tenant memberships	yes
role / permission / role_permission	role.code unique	role N-M permission; bundles = named sets	is_system, is_active	platform catalog + tenant custom (v1.1)	yes
student	id; admission_no unique per tenant	N-1 branch(current), classroom(current); 1-N guardian link; 1-N enrollment	PROVISIONAL ACTIVE INACTIVE GRADUATED TRANSFERRED WITHDRAWN ARCHIVED	tenant	yes + version
guardian	id; (tenant, phone) dedup	N-M student via student_guardian (is_primary, is_fee_payer, can_pickup, custody)	-	tenant	yes
application	id; app_no APP-YYYY-NNNN unique	N-1 lead (nullable), branch, session; 1-N documents, approvals; 1-1 admission	see Sec. 10.1	tenant	yes + version
admission_document	id	N-1 application; file_upload ref; hash dedup	PENDING VERIFIED REJECTED RESUBMIT	tenant	yes
enrollment	id; unique (student, academic_year)	N-1 student, class, fee_plan	OFFERED ACCEPTED FEE_PENDING ENROLLED	tenant	yes

Entity	PK / keys	Core relationships	Status column	Scope	Audit cols
classroom / section	id	N-1 branch, academic year, program; N-M teachers via teacher_class	-	tenant	yes
attendance	unique (student, date)	N-1 student, classroom, session	PRESENT ABSENT LATE HALF_DAY LEAVE	tenant (partitioned by year)	yes
attendance_correction	id	N-1 attendance; approver ref	PENDING APPROVED REJECTED	tenant	yes
observation / activity / milestone / milestone_assessment / portfolio_item	id	N-1 student, classroom; observer ref	observation: shared flag + ai_draft; assessment: 5 levels	tenant	yes
fee_head / fee_plan / fee_plan_item / fee_installment	id	plan N-1 program (branch optional); items N-1 head	-	tenant	yes
invoice / invoice_item	invoice_no unique per school per FY; version	N-1 student, fee_plan; 1-N items, payments via allocation	DRAFT ISSUED PARTIALLY_PAID PAID OVERDUE CANCELLED WRITTEN_OFF	tenant	yes + version
payment / payment_allocation / receipt	payment_no, receipt_no (per branch per FY)	payment N-M invoice via allocation; receipt 1-1 payment	payment: PENDING SUCCESS FAILED REFUNDED	tenant	yes
refund / credit_note	refund_no; credit_no	N-1 payment, invoice; approvals 1-N	REQUESTED APPROVED PROCESSED REJECTED CANCELLED	tenant	yes
announcement / broadcast / chat_room / chat_message	id	audiences; recipients log with delivery + read	DRAFT SCHEDULED PUBLISHED CANCELLED (announcement)	tenant	yes
notification_queue / *_log (sms/whatsapp/email/push)	id	trigger_event, template ref, provider refs	QUEUED RUNNING COMPLETED FAILED CANCELLED	tenant	yes
ai_insight / ai_usage (NEW)	id	related_entity polymorphic; reviewer ref	DRAFT ACCEPTED EDITED REJECTED	tenant	yes
audit_log	id; BRIN occurred_at	actor, entity polymorphic	-	tenant + platform mirror	append-only

27.2 Cross-Cutting Conventions

- PKs: UUID v7 for all new tables (time-sortable); composite PKs where tenant isolation demands (tenant_id, id); foreign keys RESTRICT for financial links, SET NULL for soft references (created_by).
- Audit block on every business table: created_at/by, updated_at/by, deleted_at/by (soft delete), version int; partial index WHERE deleted_at IS NULL.
- Money: bigint paise; percents numeric(5,2); timestamps timestamptz UTC; dates DATE (attendance_date); enums as PG native enums only when stable, otherwise varchar + CHECK.
- Partitioning: attendance, audit_log, notifications by tenant + month/year (ADR-051/046); invoice/report archives to cold storage per retention policy.
- Sequence counters table (per branch per FY per doc type) drives application/invoice/receipt numbering inside the issuing transaction (SELECT ... FOR UPDATE).
- The Prisma-only additions retained: credit_note, journal_entry (ledger projection), financial_year; the ERD-only tables retained: fee_installment, payment_allocation, cheque handling, late_fee_rule, lead scoring, guardians join model.

28. Edge Cases

The audit brief lists 29 edge situations that the flow must survive. The table below defines the canonical system behaviour for each, which becomes the QA acceptance baseline (Test Strategy flows reference this table). All edge-case actions are audit-logged; where a reason is marked mandatory, the API returns 422 without it.

Table 28-1: Edge case register

#	Edge case	Canonical behaviour
1	Duplicate admission (same child twice)	(branch, session, child dob+name hash) unique check -> 409 ADMISSION_DUPLICATE with link to existing application
2	Duplicate student (same child re-admitted)	Allowed as READMISSION type with new admission number; history links prior records via guardian+dob match; never a second ACTIVE enrollment in the same year
3	Duplicate parent	Guardian dedup by phone+tenant on create/link; merge tool unifies duplicates (GuardianMerge workflow); merged record becomes alias

#	Edge case	Canonical behaviour
4	Missing documents at submission	Application may submit but stays DOCUMENT_PENDING; approval action blocked (422 CHECKLIST_INCOMPLETE); window reminder cadence to parent
5	Invalid document (corrupt/wrong type)	Upload validation: MIME whitelist + magic bytes + size; verification rejection with reason -> RESUBMIT loop; OCR mismatch flags manual review
6	Rejected application	Status REJECTED + reason + feedback_to_parent + appeal_eligible; re-application creates a NEW application (no status resurrection)
7	Withdrawn application	Parent-initiated WITHDRAWN with confirmation; lead (if any) returns to NURTURE or LOST per counsellor choice
8	Cancelled admission	School-initiated (offer expiry, duplicate, fraud); cancellation reason mandatory; payments already made flow to refund pipeline
9	Student transfer (internal)	New classroom/branch assignment next day onward; attendance/fees re-scope; history TRANSFERRED event; same-day attendance rows preserved in old class
10	Student transfer (external)	TC generated, dues cleared mandatory, status TRANSFERRED; parent access revoked at effective date
11	Student promotion	Year-boundary job: next-year class assignment + history; students failing criteria flagged for Principal case review (BRC R-OPS-005 / R-ACD-011)
12	Student withdrawal	Dues settlement check (block if outstanding without Owner countersign); status WITHDRAWN; reminders stop; parent access revoked
13	Archived student in lists	Excluded by default; Show-deleted/Archive toggle for admins; archive register page lists reason + retention date
14	Inactive student in operations	INACTIVE excluded from rosters, attendance, transport and reminders; visible in student list with badge; reinstate restores

#	Edge case	Canonical behaviour
15	Suspended school	All writes 403 TENANT_SUSPENDED; reads + exports allowed; platform banner; parent app shows maintenance notice
16	Deleted user (staff)	Soft delete: sessions revoked, assignments removed, audit author preserved by id; records keep created_by meaning
17	Deactivated user mid-flow	Open tasks (approvals, corrections) auto-reassign to role queue; notification to Principal; nothing orphaned
18	Teacher reassignment mid-year	Teacher-class mapping update from effective date; past attendance/observations keep original teacher; today's unsubmitted sheets follow new mapping
19	Class reassignment (student)	Same as internal transfer but within branch; unique (student, year) constraint forces closing the previous assignment first
20	Branch transfer of staff	Assignment scope updated; multi-branch staff keep two scoped assignments; dashboards follow active branch switcher
21	Fee overdue	Reminder cadence (Table 19-2) + late-fee slabs; > 60d service hold needs Principal approval; student remains ACTIVE (welfare rule)
22	Partial payment	payment_allocation across invoices; invoice PARTIALLY_PAID; receipt for paid amount; balance recomputed (total - paid) with DB CHECK
23	Failed payment	Payment FAILED with gateway reason; invoice state unchanged; parent retry allowed; webhook/idempotency prevents double charge (SERIALIZABLE on money)
24	Refund request	Eligibility (90-day window) + tier approval (Table 24-3); processed via gateway reversal or manual transfer with proof; receipt annotated; audit every step
25	Attendance correction after lock	No hard lock: CorrectionRequest path with reason + approver; monthly summary recomputes on approval; both values audited
26	Missing parent (no guardian)	Enrollment blocked (≥ 1 guardian mandatory); legacy imports allowed with guardian_placeholder flag + Principal attestation task

#	Edge case	Canonical behaviour
27	Multiple guardians	Exactly one is _primary; fee payer and pickup flags independent; communication goes to all non-opted-out guardians; custody flags restrict as configured
28	Missing teacher (class unassigned)	Class creation requires a teacher before go-live checklist item 5; interim: Principal holds marking rights; attendance escalation path covers gaps
29	Missing fee plan	Enrollment FEE_PENDING allowed with invoice deferred; setup-progress card flags program without plan; invoice generation job skips + alerts Accountant

29. End-to-End Flow Diagrams

Every flow below follows the normative shape: START > ACTION > VALIDATION > DECISION > SUCCESS / FAILURE > NEXT STATE. These are the QA-traceable flows: the Test Strategy's ten critical E2E journeys map onto flows 8-13, 17-20 and 22 of this section. Permissions cited are from Sec. 24; audit points are from Sec. 22.

29.1 Platform & School Bootstrap

F1 PLATFORM ONBOARDING: START -> platform admin signs in -> MFA -> [role ok? yes] -> platform console -> NEXT: F2

F2 SCHOOL CREATION: START -> create tenant (5 steps) -> validate uniqueness + plan -> commit -> SUCCESS: tenant PENDING_ACTIVATION + owner invited / FAILURE: inline field errors -> NEXT: F3

F3 SCHOOL ACTIVATION: START -> KYC + signatory + payment gate -> [all true?] -> SUCCESS: ONBOARDING / FAILURE: checklist persists -> NEXT: F4

F4 SCHOOL ONBOARDING: START -> owner first login (set password, MFA) -> wizard 9 steps -> [Principal ACTIVE + year + class?] -> SUCCESS: ACTIVE + go-live / FAILURE: resume later -> NEXT: F5

F5 SCHOOL SETUP: START -> manage years/branches/classes/fees/staff -> validations per module -> SUCCESS: config saved (audit) -> NEXT: daily ops (F13+)

F6 USER CREATION: START -> authorized creator submits user+role+scope -> validate email/phone unique, role exists -> invite sent -> SUCCESS: PENDING_VERIFICATION / FAILURE: 409/422 -> NEXT: F7

F7 LOGIN: START -> credentials -> [status/tenant/role checks per Sec. 5.2] -> SUCCESS: tokens + role route / FAILURE: lockout or specific error -> NEXT: role home

29.2 Admissions To Enrollment

F8 LEAD CREATION: START -> capture (form/walk-in/campaign) -> phone dedup check -> [duplicate?] merge-offered -> SUCCESS: NEW lead + assignee (round-robin) -> NEXT: follow-ups

F9 ADMISSION (application): START -> convert lead / create app -> checklist + age + capacity validations -> review + approval chain -> SUCCESS: APPROVED + offer / FAILURE: REJECTED(reason)|WAITLISTED -> NEXT: F10

F10 STUDENT ENROLLMENT: START -> offer accepted -> enrollment tx (student + guardians) -> assignment steps -> fee gate -> SUCCESS: student ACTIVE / FAILURE: FEE_PENDING cadence -> NEXT: F12 parent

F11 PARENT ONBOARDING: START -> guardian link at enrollment -> invite -> first login + password -> SUCCESS: parent ACTIVE -> NEXT: portal

F12 CLASS ASSIGNMENT: START -> assign classroom -> capacity + age + year validations -> SUCCESS: assignment + history / FAILURE: 409 CAPACITY_FULL (waitlist) -> NEXT: rosters

29.3 Daily Operations & Learning

F13 ATTENDANCE: START -> teacher opens class-day -> roster loads -> mark + submit -> [duplicates? blocked] -> SUCCESS: rows + absence alerts -> corrections per Sec. 15 -> reports

F14 ACTIVITY RECORDING: START -> teacher logs activity + photos -> consent + size validations -> SUCCESS: timeline entry (<= 5 min) / FAILURE: consent-blocked flagged -> NEXT: F15

F15 OBSERVATION: START -> teacher writes (or AI draft) -> quality check (>= 20 chars) -> publish/share toggle -> SUCCESS: observation + parent notify / FAILURE: quality flag -> coaching

F16 GROWTH PASSPORT: START -> term compile job -> projections gathered -> teacher+principal review -> SUCCESS: passport PDF published / FAILURE: review notes -> revise

29.4 Finance

F17 FEE ASSIGNMENT: START -> plan assigned at enrollment/concession change -> approval tier check -> SUCCESS: schedule generated -> NEXT: F18

F18 INVOICE: START -> cycle job or manual generate -> plan + concessions computed -> DRAFT -> issue -> SUCCESS: ISSUED + D-7 reminder armed / FAILURE: alert Accountant (no plan)

F19 PAYMENT: START -> parent pays online OR office records -> Idempotency-Key + gateway -> [success?] -> SUCCESS: allocations + receipt <= 60s + events / FAILURE: FAILED + retry path

F20 RECEIPT: START -> payment SUCCESS -> number per branch counter -> PDF stored -> SUCCESS: receipt issued + notify (reprint = copy, audited)

29.5 Communication, Insight & Reporting

F21 COMMUNICATION: START -> compose (chat/announcement) -> audience-permission matrix + quiet hours -> [approval needed? tier] -> SUCCESS: queued -> sent -> delivered -> read (audit)

F22 REPORTS: START -> user opens report -> scope + permission check -> query (replica/MV) -> SUCCESS: view/export (audit) / FAILURE: 403 or empty-state guidance

F23 AI RECOMMENDATION: START -> feature trigger (draft/alert) -> gateway quota + consent + PII gate -> SUCCESS: ai_insight DRAFT -> human review -> ACCEPT/EDIT/REJECT (audit) / FAILURE: quota/consent block

29.6 Student & Staff Exits, Platform Actions

F24 STUDENT TRANSFER: START -> transfer request -> [internal?] branch rescope : TC + dues check -> SUCCESS: status/history per Sec. 12 / FAILURE: dues block (countersign path)

F25 STUDENT PROMOTION: START -> year-end job -> criteria eval (attendance >= 75%, milestones, fees) -> [eligible?] auto-assign next class / flag case review -> SUCCESS: history event

F26 STUDENT WITHDRAWAL: START -> request -> dues settlement -> confirm -> SUCCESS: WITHDRAWN + access revoked / FAILURE: outstanding dues blocked

F27 STUDENT ARCHIVE: START -> retention job / admin action -> reason + DPDP minimize -> SUCCESS: ARCHIVED read-only -> later purge [IRREVERSIBLE]

F28 USER DEACTIVATION: START -> admin action -> task reassignment -> sessions revoked -> SUCCESS: INACTIVE (reactivable) / audited

F29 SCHOOL SUSPENSION: START -> trigger (dunning/violation/manual) -> writes blocked tenant-wide -> SUCCESS: SUSPENDED + notice / resolve -> ACTIVE (audit pair)

30. FINAL CANONICAL PREONE FLOW

This is the one master flow that survives the audit. It corrects the draft sequence in the brief in two places: school onboarding and setup precede the admission CRM being staffed (a school cannot run admissions without at least a Principal and a fee structure), and AI insights are positioned strictly after validated operational data exists, feeding back into operations through human-approved actions only.

```

PREONE PLATFORM
  v
PLATFORM ADMIN (console) [F1]
  v
SCHOOL TENANT CREATED (DRAFT -> PENDING_ACTIVATION) [F2]
  v
SCHOOL ACTIVATION (KYC + subscription gate) [F3]
  v
OWNER FIRST LOGIN -> SCHOOL ONBOARDING (wizard) [F4]
  v
SCHOOL SETUP (years, branches, classes, fees, staff, roles) [F5]
  v
USERS + RBAC LIVE (staff ACTIVE with scoped bundles) [F6][F7]
  v
SCHOOL LIVE: ADMISSIONS CRM (leads, follow-ups, visits, funnel) [F8]
  v

```

```

APPLICATION -> DOCUMENT VERIFICATION -> APPROVAL CHAIN    [F9]
  v
ENROLLMENT (student + guardians + assignments + fee gate)    [F10][F11][F12]
  v
DAILY SCHOOL OPERATIONS
  |- ATTENDANCE (mark -> correct -> summaries)                [F13]
  |- ACTIVITIES + OBSERVATIONS (consent, quality, sharing)    [F14][F15]
  |- GROWTH PASSPORT (projections -> term report)             [F16]
  v
FEES & FINANCE (plans -> invoices -> payments -> receipts -> refunds) [F17]
[F18][F19][F20]
  v
COMMUNICATION (chat, broadcasts, event notifications)        [F21]
  v
REPORTS (operational, academic, financial, compliance)        [F22]
  v
AI INSIGHTS (validated data -> drafts/advisories -> HUMAN review -> action)
[F23]
  v
LIFECYCLE EXITS (transfer, promotion, withdrawal, archive; user deactivation;
school suspension) [F24..F29]
  v
AUDIT / SYSTEM MONITORING (every step above; hash-chained; 7-year retention;
SIEM mirror)

```

Read as a contract: every arrow above is a validated API transition with a permission, an audit record and a defined failure state; every module owns its tables; every event crosses domains only through the outbox; and the human is the final authority wherever child data, money or communication leaves the system.

31. Open Decisions / Items Requiring Product Approval

These are genuine product choices that the sources do not settle. Each is registered with options, this document's recommendation, the affected canonical sections, and the team best placed to own the sign-off. Until decided, the recommendation stands as the working canonical behaviour (clearly marked RECOMMENDED in the body), so development is never blocked.

Table 31-1: Decision register

ID	Decision	Options	Recommendation	Owner / impact
OD-1	Growth Passport v1.0 composition	(a) As proposed in Sec. 17 (projection view + term PDF); (b) defer term; (c) standalone entity	(a) projection composition; no new silo	Product + UX; sections 16, 17, 23

ID	Decision	Options	Recommendation	Owner / impact
OD-2	Transport in v1.0 scope	(a) Feature-flagged out (PRD); (b) ship with DDD context	(a) out of v1.0, schema retained; TRANSPORT_MANAGER role ships dormant	Product; sections 03, 04, 19
OD-3	Observation frequency target	(a) min 1/target 2; (b) hard 3 (PRD)	(a) min 1 enforced, target 2 on dashboards	Product + Academic; section 16
OD-4	Refund eligibility window	(a) 90 days; (b) 30 days (DDD service); (c) none	(a) 90 days, configurable per tenant	Product + Finance; section 18
OD-5	Invoice numbering scope	(a) per school per FY; (b) per branch per FY (BRC)	(a) per school per FY; receipts per branch	Finance compliance; section 18, 27
OD-6	AI provider set	(a) Z.ai GLM + OpenAI reserve; (b) add Anthropic (DDD ACL)	(a) two-provider ACL now; provider abstraction ready for (b)	CTO; section 20
OD-7	Parent MFA factors	(a) SMS OTP allowed; (b) TOTP/email only	(a) SMS OTP for parents (usability), TOTP for staff	Product + Security; section 05
OD-8	Finance role naming	(a) ACCOUNTANT; (b) Finance User (Security)	(a) ACCOUNTANT (matches data model + market usage)	Product; section 03
OD-9	BRC rule-ID reissue	(a) re-map DDD citations to BRC IDs; (b) dual registry	(a) BRC v1.1 with reconciliation appendix	Product + Architecture; section 02
OD-10	Custom roles per tenant in v1.0?	(a) v1.0; (b) v1.1 (system roles only)	(b) v1.1 using ADR-064 catalog + no-escalation guard	CTO + Product; section 24
OD-11	GST configuration surface	(a) fixed BRC defaults; (b) per-tenant editable tax matrix	(a) defaults now, (b) behind settings flag	Finance + Product; section 18

32. Development Readiness Checklist

The checklist below converts this document into team-level acceptance gates. An item is done when the referenced canonical section is implemented, tested (Test Strategy mapping) and audited. QA treats Sec. 25 state machines and Sec. 28 edge cases as the coverage baseline; DevOps items reference ADR-111, which remains valid.

Table 32-1: CTO / Tech Lead gates

#	Item	Done when
T1	Conflict register adopted	All rulings C01-C34 accepted; superseded doc sections annotated or docs re-issued

#	Item	Done when
T2	Stack freeze re-confirmed	NestJS + Prisma + PostgreSQL 16 + Redis + BullMQ + Next.js 16 declared; ADR-020 Java marked superseded
T3	Open decisions OD-1..OD-11 dispositioned	Each has an owner and a date; recommendations active meanwhile
T4	Isolation test suite	Cross-tenant probe suite green at API, service, repo and raw-SQL layers (Test Strategy 11.8.2)
T5	Audit completeness	Linters fail any state-changing route without audit decorator; hash-chain verification job scheduled

Table 32-2: Product / UX gates

#	Item	Done when
P1	IA approved	Sec. 23 tree reviewed; nav config generated from permission metadata
P2	Page specs complete	Sec. 23.2 extended to all module pages with the Student-Profile benchmark template
P3	Onboarding wizard spec	Sec. 07 steps, gates and skip rules signed off; resumability designed
P4	Growth Passport decision	OD-1 closed; passport surfaces designed per outcome
P5	Notification catalog	Sec. 19.3 mapped to templates per channel and language

Table 32-3: Backend gates

#	Item	Done when
B1	Canonical schema migrated	Sec. 27 catalog implemented; two-generation merge complete; pause everywhere; uuid7 for new tables
B2	State machines enforced	Sec. 25 transitions in domain layer; invalid transitions 409 INVALID_STATE_TRANSITION
B3	RBAC engine	module:action grammar; scoped assignments; effective-permission cache with 60s freshness; matrix tests green

#	Item	Done when
B4	Admission pipeline + enrollment saga	Sec. 10/11 implemented with five status domains and workflow SLAs
B5	Attendance correction flow	Same-day edit + CorrectionRequest approval path + monthly recompute (Sec. 15)
B6	Finance rails	Immutable invoices, allocations, receipts counters, refund tiers, late-fee slabs, GST config (Sec. 18)
B7	Notification engine	Sec. 19 pipeline with rate limits, quiet hours, DLQ; trigger catalog live
B8	AI gateway + persistence	ai_insight/ai_usage tables; provenance; never-write guard tests (Sec. 20)
B9	Audit pipeline	Superset record (Sec. 22); same-tx writes; platform + tenant query APIs
B10	API contract regenerated	OpenAPI re-issued from the catalog with canonical envelope, paths, pagination (Sec. 26)

Table 32-4: Frontend gates

#	Item	Done when
F1	Shell + navigation	Sec. 23 IA implemented; permission-filtered menus; mobile bottom nav; teacher tablet home
F2	Route guards	Edge middleware + layout guards per Sec. 05; 403 page with reason; no silent redirects
F3	State standards	Skeleton loading, empty CTAs, error toasts with traceId; optimistic patterns per FA Sec. 20
F4	Core pages	Sec. 23.2 pages implemented with actions, validations and states as specified
F5	Onboarding + platform console	Wizard and console per Sec. 06/07; impersonation banner

Table 32-5: QA gates

#	Item	Done when
Q1	State-machine coverage	Every Sec. 25 transition has positive + negative tests; irreversible transitions proven immutable
Q2	RBAC matrix tests	Sec. 24.3 matrix encoded as test data; role x endpoint allow/deny green; escalation attempts 403
Q3	Edge-case suite	All 29 Sec. 28 rows have automated or scripted checks
Q4	E2E critical flows	10 critical journeys (Sec. 29 mapping) automated; admission TAT < 7 days simulated
Q5	Finance integrity	Idempotency replay, double-charge race, allocation math, receipt uniqueness, refund tiers tested

Table 32-6: DevOps gates

#	Item	Done when
D1	Environments	Six-env topology per ADR-111; prod access break-glass with MFA + audit
D2	Secrets decision	OD on Vault vs cloud secret manager closed; rotation schedule per Security Sec. 13
D3	Retention + purge jobs	Retention job per ADR-053 with legal-hold support; purge certificates produced
D4	Audit SIEM mirror	Outbox -> SIEM stream live; hash-chain verify + alert scheduled
D5	Backup/DR	RPO <= 15 min / RTO per ADR-111; quarterly restore drills scheduled

With these gates green, PreOne has one flow, one vocabulary, one permission model and one audit trail - and every team can answer, without returning to the conflicting sources: what screen comes next, which action is allowed, which API is called, which data changes, which permission is required, which status changes, what happens on failure, and what is logged.